

## 12.2 Distribution tables

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**Table A1. Table of Uniform(0,1) random numbers**

| $z$ | 1     | 2     | 3     | 4     | 5     | 6     | 7     | 8     | 9     | 10    |
|-----|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1   | .9501 | .8381 | .7948 | .4154 | .6085 | .4398 | .2974 | .7165 | .7327 | .8121 |
| 2   | .2311 | .0196 | .9568 | .3050 | .0158 | .3400 | .0492 | .5113 | .4222 | .6101 |
| 3   | .6068 | .6813 | .5226 | .8744 | .0164 | .3142 | .6932 | .7764 | .9614 | .7015 |
| 4   | .4860 | .3795 | .8801 | .0150 | .1901 | .3651 | .6501 | .4893 | .0721 | .0922 |
| 5   | .8913 | .8318 | .1730 | .7680 | .5869 | .3932 | .9830 | .1859 | .5534 | .4249 |
| 6   | .7621 | .5028 | .9797 | .9708 | .0576 | .5915 | .5527 | .7006 | .2920 | .3756 |
| 7   | .4565 | .7095 | .2714 | .9901 | .3676 | .1197 | .4001 | .9827 | .8580 | .1662 |
| 8   | .0185 | .4289 | .2523 | .7889 | .6315 | .0381 | .1988 | .8066 | .3358 | .8332 |
| 9   | .8214 | .3046 | .8757 | .4387 | .7176 | .4586 | .6252 | .7036 | .6802 | .8386 |
| 10  | .4447 | .1897 | .7373 | .4983 | .6927 | .8699 | .7334 | .4850 | .0534 | .4516 |
| 11  | .6154 | .1934 | .1365 | .2140 | .0841 | .9342 | .3759 | .1146 | .3567 | .9566 |
| 12  | .7919 | .6822 | .0118 | .6435 | .4544 | .2644 | .0099 | .6649 | .4983 | .1472 |
| 13  | .9218 | .3028 | .8939 | .3200 | .4418 | .1603 | .4199 | .3654 | .4344 | .8699 |
| 14  | .7382 | .5417 | .1991 | .9601 | .3533 | .8729 | .7537 | .1400 | .5625 | .7694 |
| 15  | .1763 | .1509 | .2987 | .7266 | .1536 | .2379 | .7939 | .5668 | .6166 | .4442 |
| 16  | .4057 | .6979 | .6614 | .4120 | .6756 | .6458 | .9200 | .8230 | .1133 | .6206 |
| 17  | .9355 | .3784 | .2844 | .7446 | .6992 | .9669 | .8447 | .6739 | .8983 | .9517 |
| 18  | .9169 | .8600 | .4692 | .2679 | .7275 | .6649 | .3678 | .9994 | .7546 | .6400 |
| 19  | .4103 | .8537 | .0648 | .4399 | .4784 | .8704 | .6208 | .9616 | .7911 | .2473 |
| 20  | .8936 | .5936 | .9883 | .9334 | .5548 | .0099 | .7313 | .0589 | .8150 | .3527 |
| 21  | .0579 | .4966 | .5828 | .6833 | .1210 | .1370 | .1939 | .3603 | .6700 | .1879 |
| 22  | .3529 | .8998 | .4235 | .2126 | .4508 | .8188 | .9048 | .5485 | .2009 | .4906 |
| 23  | .8132 | .8216 | .5155 | .8392 | .7159 | .4302 | .5692 | .2618 | .2731 | .4093 |
| 24  | .0099 | .6449 | .3340 | .6288 | .8928 | .8903 | .6318 | .5973 | .6262 | .4635 |
| 25  | .1389 | .8180 | .4329 | .1338 | .2731 | .7349 | .2344 | .0493 | .5369 | .6109 |
| 26  | .2028 | .6602 | .2259 | .2071 | .2548 | .6873 | .5488 | .5711 | .0595 | .0712 |
| 27  | .1987 | .3420 | .5798 | .6072 | .8656 | .3461 | .9316 | .7009 | .0890 | .3143 |
| 28  | .6038 | .2897 | .7604 | .6299 | .2324 | .1660 | .3352 | .9623 | .2713 | .6084 |
| 29  | .2722 | .3412 | .5298 | .3705 | .8049 | .1556 | .6555 | .7505 | .4091 | .1750 |
| 30  | .1988 | .5341 | .6405 | .5751 | .9084 | .1911 | .3919 | .7400 | .4740 | .6210 |
| 31  | .0153 | .7271 | .2091 | .4514 | .2319 | .4225 | .6273 | .4319 | .9090 | .2460 |
| 32  | .7468 | .3093 | .3798 | .0439 | .2393 | .8560 | .6991 | .6343 | .5962 | .5874 |
| 33  | .4451 | .8385 | .7833 | .0272 | .0498 | .4902 | .3972 | .8030 | .3290 | .5061 |
| 34  | .9318 | .5681 | .6808 | .3127 | .0784 | .8159 | .4136 | .0839 | .4782 | .4648 |
| 35  | .4660 | .3704 | .4611 | .0129 | .6408 | .4608 | .6552 | .9455 | .5972 | .5414 |
| 36  | .4186 | .7027 | .5678 | .3840 | .1909 | .4574 | .8376 | .9159 | .1614 | .9423 |
| 37  | .8462 | .5466 | .7942 | .6831 | .8439 | .4507 | .3716 | .6020 | .8295 | .3418 |
| 38  | .5252 | .4449 | .0592 | .0928 | .1739 | .4122 | .4253 | .2536 | .9561 | .4018 |
| 39  | .2026 | .6946 | .6029 | .0353 | .1708 | .9016 | .5947 | .8735 | .5955 | .3077 |
| 40  | .6721 | .6213 | .0503 | .6124 | .9943 | .0056 | .5657 | .5134 | .0287 | .4116 |

Table A2. Binomial distribution

$$F(x) = \mathbf{P}\{X \leq x\} = \sum_{k=0}^x \binom{n}{k} p^k (1-p)^{n-k}$$

| <i>n</i> | <i>x</i> | <i>p</i> |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
|----------|----------|----------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
|          |          | .050     | .100 | .150 | .200 | .250 | .300 | .350 | .400 | .450 | .500 | .550 | .600 | .650 | .700 | .750 | .800 | .850 | .900 | .950 |
| 1        | 0        | .950     | .900 | .850 | .800 | .750 | .700 | .650 | .600 | .550 | .500 | .450 | .400 | .350 | .300 | .250 | .200 | .150 | .100 | .050 |
| 2        | 0        | .903     | .810 | .723 | .640 | .563 | .490 | .423 | .360 | .303 | .250 | .203 | .160 | .123 | .090 | .063 | .040 | .023 | .010 | .003 |
|          | 1        | .998     | .990 | .978 | .960 | .938 | .910 | .878 | .840 | .798 | .750 | .698 | .640 | .578 | .510 | .438 | .360 | .278 | .190 | .098 |
| 3        | 0        | .857     | .729 | .614 | .512 | .422 | .343 | .275 | .216 | .166 | .125 | .091 | .064 | .043 | .027 | .016 | .008 | .003 | .001 | .000 |
|          | 1        | .993     | .972 | .939 | .896 | .844 | .784 | .718 | .648 | .575 | .500 | .425 | .352 | .282 | .216 | .156 | .104 | .061 | .028 | .007 |
|          | 2        | 1.0      | .999 | .997 | .992 | .984 | .973 | .957 | .936 | .909 | .875 | .834 | .784 | .725 | .657 | .578 | .488 | .386 | .271 | .143 |
| 4        | 0        | .815     | .656 | .522 | .410 | .316 | .240 | .179 | .130 | .092 | .063 | .041 | .026 | .015 | .008 | .004 | .002 | .001 | .000 | .000 |
|          | 1        | .986     | .948 | .890 | .819 | .738 | .652 | .563 | .475 | .391 | .313 | .241 | .179 | .126 | .084 | .051 | .027 | .012 | .004 | .000 |
|          | 2        | 1.0      | .996 | .988 | .973 | .949 | .916 | .874 | .821 | .759 | .688 | .609 | .525 | .437 | .348 | .262 | .181 | .110 | .052 | .014 |
|          | 3        | 1.0      | 1.0  | .999 | .998 | .996 | .992 | .985 | .974 | .959 | .938 | .908 | .870 | .821 | .760 | .684 | .590 | .478 | .344 | .185 |
| 5        | 0        | .774     | .590 | .444 | .328 | .237 | .168 | .116 | .078 | .050 | .031 | .018 | .010 | .005 | .002 | .001 | .000 | .000 | .000 | .000 |
|          | 1        | .977     | .919 | .835 | .737 | .633 | .528 | .428 | .337 | .256 | .188 | .131 | .087 | .054 | .031 | .016 | .007 | .002 | .000 | .000 |
|          | 2        | .999     | .991 | .973 | .942 | .896 | .837 | .765 | .683 | .593 | .500 | .407 | .317 | .235 | .163 | .104 | .058 | .027 | .009 | .001 |
|          | 3        | 1.0      | 1.0  | .998 | .993 | .984 | .969 | .946 | .913 | .869 | .813 | .744 | .663 | .572 | .472 | .367 | .263 | .165 | .081 | .023 |
|          | 4        | 1.0      | 1.0  | 1.0  | 1.0  | .999 | .998 | .995 | .990 | .982 | .969 | .950 | .922 | .884 | .832 | .763 | .672 | .556 | .410 | .226 |
| 6        | 0        | .735     | .531 | .377 | .262 | .178 | .118 | .075 | .047 | .028 | .016 | .008 | .004 | .002 | .001 | .000 | .000 | .000 | .000 | .000 |
|          | 1        | .967     | .886 | .776 | .655 | .534 | .420 | .319 | .233 | .164 | .109 | .069 | .041 | .022 | .011 | .005 | .002 | .000 | .000 | .000 |
|          | 2        | .998     | .984 | .953 | .901 | .831 | .744 | .647 | .544 | .442 | .344 | .255 | .179 | .117 | .070 | .038 | .017 | .006 | .001 | .000 |
|          | 3        | 1.0      | .999 | .994 | .983 | .962 | .930 | .883 | .821 | .745 | .656 | .558 | .456 | .353 | .256 | .169 | .099 | .047 | .016 | .002 |
|          | 4        | 1.0      | 1.0  | 1.0  | .998 | .995 | .989 | .978 | .959 | .931 | .891 | .836 | .767 | .681 | .580 | .466 | .345 | .224 | .114 | .033 |
|          | 5        | 1.0      | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .998 | .996 | .992 | .984 | .972 | .953 | .925 | .882 | .822 | .738 | .623 | .469 | .265 |
| 7        | 0        | .698     | .478 | .321 | .210 | .133 | .082 | .049 | .028 | .015 | .008 | .004 | .002 | .001 | .000 | .000 | .000 | .000 | .000 | .000 |
|          | 1        | .956     | .850 | .717 | .577 | .445 | .329 | .234 | .159 | .102 | .063 | .036 | .019 | .009 | .004 | .001 | .000 | .000 | .000 | .000 |
|          | 2        | .996     | .974 | .926 | .852 | .756 | .647 | .532 | .420 | .316 | .227 | .153 | .096 | .056 | .029 | .013 | .005 | .001 | .000 | .000 |
|          | 3        | 1.0      | .997 | .988 | .967 | .929 | .874 | .800 | .710 | .608 | .500 | .392 | .290 | .200 | .126 | .071 | .033 | .012 | .003 | .000 |
|          | 4        | 1.0      | 1.0  | .999 | .995 | .987 | .971 | .944 | .904 | .847 | .773 | .684 | .580 | .468 | .353 | .244 | .148 | .074 | .026 | .004 |
|          | 5        | 1.0      | 1.0  | 1.0  | 1.0  | .999 | .996 | .991 | .981 | .964 | .938 | .898 | .841 | .766 | .671 | .555 | .423 | .283 | .150 | .044 |
|          | 6        | 1.0      | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .998 | .996 | .992 | .985 | .972 | .951 | .918 | .867 | .790 | .679 | .522 | .302 |
| 8        | 0        | .663     | .430 | .272 | .168 | .100 | .058 | .032 | .017 | .008 | .004 | .002 | .001 | .000 | .000 | .000 | .000 | .000 | .000 | .000 |
|          | 1        | .943     | .813 | .657 | .503 | .367 | .255 | .169 | .106 | .063 | .035 | .018 | .009 | .004 | .001 | .000 | .000 | .000 | .000 | .000 |
|          | 2        | .994     | .962 | .895 | .797 | .679 | .552 | .428 | .315 | .220 | .145 | .088 | .050 | .025 | .011 | .004 | .001 | .000 | .000 | .000 |
|          | 3        | 1.0      | .995 | .979 | .944 | .886 | .806 | .706 | .594 | .477 | .363 | .260 | .174 | .106 | .058 | .027 | .010 | .003 | .000 | .000 |
|          | 4        | 1.0      | 1.0  | .997 | .990 | .973 | .942 | .894 | .826 | .740 | .637 | .523 | .406 | .294 | .194 | .114 | .056 | .021 | .005 | .000 |
|          | 5        | 1.0      | 1.0  | 1.0  | 1.0  | .999 | .996 | .989 | .975 | .950 | .912 | .855 | .780 | .685 | .572 | .448 | .321 | .203 | .105 | .038 |
|          | 6        | 1.0      | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .996 | .991 | .982 | .965 | .937 | .894 | .831 | .745 | .633 | .497 | .343 | .187 | .057 |
|          | 7        | 1.0      | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .998 | .996 | .992 | .983 | .968 | .942 | .900 | .832 | .728 | .570 | .337 |
| 9        | 0        | .630     | .387 | .232 | .134 | .075 | .040 | .021 | .010 | .005 | .002 | .001 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 |
|          | 1        | .929     | .775 | .599 | .436 | .300 | .196 | .121 | .071 | .039 | .020 | .009 | .004 | .001 | .000 | .000 | .000 | .000 | .000 | .000 |
|          | 2        | .992     | .947 | .859 | .738 | .601 | .463 | .337 | .232 | .150 | .090 | .050 | .025 | .011 | .004 | .001 | .000 | .000 | .000 | .000 |
|          | 3        | .999     | .992 | .966 | .914 | .834 | .730 | .609 | .483 | .361 | .254 | .166 | .099 | .054 | .025 | .010 | .003 | .001 | .000 | .000 |
|          | 4        | 1.0      | .999 | .994 | .980 | .951 | .901 | .828 | .733 | .621 | .500 | .379 | .267 | .172 | .099 | .049 | .020 | .006 | .001 | .000 |
|          | 5        | 1.0      | 1.0  | .999 | .997 | .990 | .975 | .946 | .901 | .834 | .746 | .639 | .517 | .391 | .270 | .166 | .086 | .034 | .008 | .001 |
|          | 6        | 1.0      | 1.0  | 1.0  | 1.0  | .999 | .996 | .989 | .975 | .950 | .910 | .850 | .768 | .663 | .537 | .399 | .262 | .141 | .053 | .008 |
|          | 7        | 1.0      | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .996 | .991 | .980 | .961 | .929 | .879 | .804 | .700 | .564 | .401 | .225 | .071 |
|          | 8        | 1.0      | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .998 | .995 | .990 | .979 | .960 | .925 | .866 | .768 | .613 | .370 |
| 10       | 0        | .599     | .349 | .197 | .107 | .056 | .028 | .013 | .006 | .003 | .001 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 |
|          | 1        | .914     | .736 | .544 | .376 | .244 | .149 | .086 | .046 | .023 | .011 | .005 | .002 | .001 | .000 | .000 | .000 | .000 | .000 | .000 |
|          | 2        | .988     | .930 | .820 | .678 | .526 | .383 | .262 | .167 | .100 | .055 | .027 | .012 | .005 | .002 | .000 | .000 | .000 | .000 | .000 |
|          | 3        | .999     | .987 | .950 | .879 | .776 | .650 | .514 | .382 | .266 | .172 | .102 | .055 | .026 | .011 | .004 | .001 | .000 | .000 | .000 |
|          | 4        | 1.0      | .998 | .990 | .967 | .922 | .850 | .751 | .633 | .504 | .377 | .262 | .166 | .095 | .047 | .020 | .006 | .001 | .000 | .000 |
|          | 5        | 1.0      | 1.0  | .999 | .994 | .980 | .953 | .905 | .834 | .738 | .623 | .496 | .367 | .249 | .150 | .078 | .033 | .010 | .002 | .000 |
|          | 6        | 1.0      | 1.0  | 1.0  | .999 | .996 | .989 | .974 | .945 | .898 | .828 | .734 | .618 | .486 | .350 | .224 | .121 | .050 | .013 | .001 |
|          | 7        | 1.0      | 1.0  | 1.0  | 1.0  | 1.0  | .998 | .995 | .988 | .973 | .945 | .900 | .833 | .738 | .617 | .474 | .322 | .180 | .070 | .012 |
|          | 8        | 1.0      | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .998 | .995 | .989 | .977 | .954 | .914 | .851 | .756 | .624 | .456 | .264 | .086 |
|          | 9        | 1.0      | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .997 | .994 | .987 | .972 | .944 | .893 | .803 | .651 | .401 |

Table A2, continued. Binomial distribution

| $n$ | $x$ | $p$  |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
|-----|-----|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
|     |     | .050 | .100 | .150 | .200 | .250 | .300 | .350 | .400 | .450 | .500 | .550 | .600 | .650 | .700 | .750 | .800 | .850 | .900 | .950 |
| 11  | 0   | .569 | .314 | .167 | .086 | .042 | .020 | .009 | .004 | .001 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 |
|     | 1   | .898 | .697 | .492 | .322 | .197 | .113 | .061 | .030 | .014 | .006 | .002 | .001 | .000 | .000 | .000 | .000 | .000 | .000 | .000 |
|     | 2   | .985 | .910 | .779 | .617 | .455 | .313 | .200 | .119 | .065 | .033 | .015 | .006 | .002 | .001 | .000 | .000 | .000 | .000 | .000 |
|     | 3   | .998 | .981 | .931 | .839 | .713 | .570 | .426 | .296 | .191 | .113 | .061 | .029 | .012 | .004 | .001 | .000 | .000 | .000 | .000 |
|     | 4   | 1.0  | .997 | .984 | .950 | .885 | .790 | .668 | .533 | .397 | .274 | .174 | .099 | .050 | .022 | .008 | .002 | .000 | .000 | .000 |
|     | 5   | 1.0  | 1.0  | .997 | .988 | .966 | .922 | .851 | .753 | .633 | .500 | .367 | .247 | .149 | .078 | .034 | .012 | .003 | .000 | .000 |
|     | 6   | 1.0  | 1.0  | 1.0  | .998 | .992 | .978 | .950 | .901 | .826 | .726 | .603 | .467 | .332 | .210 | .115 | .050 | .016 | .003 | .000 |
|     | 7   | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .996 | .988 | .971 | .939 | .887 | .809 | .704 | .574 | .430 | .287 | .161 | .069 | .019 | .002 |
|     | 8   | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .998 | .994 | .985 | .967 | .935 | .881 | .800 | .687 | .545 | .383 | .221 | .090 | .015 |
|     | 9   | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .998 | .994 | .986 | .970 | .939 | .887 | .803 | .678 | .508 | .303 | .102 |
| 10  | 1.0 | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .996 | .991 | .980 | .958 | .914 | .833 | .686 | .431 |      |
| 12  | 0   | .540 | .282 | .142 | .069 | .032 | .014 | .006 | .002 | .001 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 |
|     | 1   | .882 | .659 | .443 | .275 | .158 | .085 | .042 | .020 | .008 | .003 | .001 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 |
|     | 2   | .980 | .889 | .736 | .558 | .391 | .253 | .151 | .083 | .042 | .019 | .008 | .003 | .001 | .000 | .000 | .000 | .000 | .000 | .000 |
|     | 3   | .998 | .974 | .908 | .795 | .649 | .493 | .347 | .225 | .134 | .073 | .036 | .015 | .006 | .002 | .000 | .000 | .000 | .000 | .000 |
|     | 4   | 1.0  | .996 | .976 | .927 | .842 | .724 | .583 | .438 | .304 | .194 | .112 | .057 | .026 | .009 | .003 | .001 | .000 | .000 | .000 |
|     | 5   | 1.0  | .999 | .995 | .981 | .946 | .882 | .787 | .665 | .527 | .387 | .261 | .158 | .085 | .039 | .014 | .004 | .001 | .000 | .000 |
|     | 6   | 1.0  | 1.0  | .999 | .996 | .986 | .961 | .915 | .842 | .739 | .613 | .473 | .335 | .213 | .118 | .054 | .019 | .005 | .001 | .000 |
|     | 7   | 1.0  | 1.0  | 1.0  | .999 | .997 | .991 | .974 | .943 | .888 | .806 | .696 | .562 | .417 | .276 | .158 | .073 | .024 | .004 | .000 |
|     | 8   | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .998 | .994 | .985 | .964 | .927 | .866 | .775 | .653 | .507 | .351 | .205 | .092 | .026 | .002 |
|     | 9   | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .997 | .992 | .981 | .958 | .917 | .849 | .747 | .609 | .442 | .264 | .111 | .020 |
|     | 10  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .997 | .992 | .980 | .958 | .915 | .842 | .725 | .557 | .341 | .118 |
|     | 11  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .998 | .994 | .986 | .968 | .931 | .858 | .718 | .460 |
| 13  | 0   | .513 | .254 | .121 | .055 | .024 | .010 | .004 | .001 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 |
|     | 1   | .865 | .621 | .398 | .234 | .127 | .064 | .030 | .013 | .005 | .002 | .001 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 |
|     | 2   | .975 | .866 | .692 | .502 | .333 | .202 | .113 | .058 | .027 | .011 | .004 | .001 | .000 | .000 | .000 | .000 | .000 | .000 | .000 |
|     | 3   | .997 | .966 | .882 | .747 | .584 | .421 | .278 | .169 | .093 | .046 | .020 | .008 | .003 | .001 | .000 | .000 | .000 | .000 | .000 |
|     | 4   | 1.0  | .994 | .966 | .901 | .794 | .654 | .501 | .353 | .228 | .133 | .070 | .032 | .013 | .004 | .001 | .000 | .000 | .000 | .000 |
|     | 5   | 1.0  | .999 | .992 | .970 | .920 | .835 | .716 | .574 | .427 | .291 | .179 | .098 | .046 | .018 | .006 | .001 | .000 | .000 | .000 |
|     | 6   | 1.0  | 1.0  | .999 | .993 | .976 | .938 | .871 | .771 | .644 | .500 | .356 | .229 | .129 | .062 | .024 | .007 | .001 | .000 | .000 |
|     | 7   | 1.0  | 1.0  | 1.0  | .999 | .994 | .982 | .954 | .902 | .821 | .709 | .573 | .426 | .284 | .165 | .080 | .030 | .008 | .001 | .000 |
|     | 8   | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .996 | .987 | .968 | .930 | .867 | .772 | .647 | .499 | .346 | .206 | .099 | .034 | .006 | .000 |
|     | 9   | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .997 | .992 | .980 | .954 | .907 | .831 | .722 | .579 | .416 | .253 | .118 | .034 | .003 |
|     | 10  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .996 | .989 | .973 | .942 | .887 | .798 | .667 | .498 | .308 | .134 | .025 |
|     | 11  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .998 | .995 | .987 | .970 | .936 | .873 | .766 | .602 | .379 | .135 |
|     | 12  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .996 | .990 | .976 | .945 | .879 | .746 | .487 |
| 14  | 0   | .488 | .229 | .103 | .044 | .018 | .007 | .002 | .001 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 |
|     | 1   | .847 | .585 | .357 | .198 | .101 | .047 | .021 | .008 | .003 | .001 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 |
|     | 2   | .970 | .842 | .648 | .448 | .281 | .161 | .084 | .040 | .017 | .006 | .002 | .001 | .000 | .000 | .000 | .000 | .000 | .000 | .000 |
|     | 3   | .996 | .956 | .853 | .698 | .521 | .355 | .220 | .124 | .063 | .029 | .011 | .004 | .001 | .000 | .000 | .000 | .000 | .000 | .000 |
|     | 4   | 1.0  | .991 | .953 | .870 | .742 | .584 | .423 | .279 | .167 | .090 | .043 | .018 | .006 | .002 | .000 | .000 | .000 | .000 | .000 |
|     | 5   | 1.0  | .999 | .988 | .956 | .888 | .781 | .641 | .486 | .337 | .212 | .119 | .058 | .024 | .008 | .002 | .000 | .000 | .000 | .000 |
|     | 6   | 1.0  | 1.0  | .998 | .988 | .962 | .907 | .816 | .692 | .546 | .395 | .259 | .150 | .075 | .031 | .010 | .002 | .000 | .000 | .000 |
|     | 7   | 1.0  | 1.0  | 1.0  | .998 | .990 | .969 | .925 | .850 | .741 | .605 | .454 | .308 | .184 | .093 | .038 | .012 | .002 | .000 | .000 |
|     | 8   | 1.0  | 1.0  | 1.0  | 1.0  | .998 | .992 | .976 | .942 | .881 | .788 | .663 | .514 | .359 | .219 | .112 | .044 | .012 | .001 | .000 |
|     | 9   | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .998 | .994 | .982 | .957 | .910 | .833 | .721 | .577 | .416 | .258 | .130 | .047 | .009 | .000 |
|     | 10  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .996 | .989 | .971 | .937 | .876 | .780 | .645 | .479 | .302 | .147 | .044 | .004 |
|     | 11  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .998 | .994 | .983 | .960 | .916 | .839 | .719 | .552 | .352 | .158 | .030 |
|     | 12  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .997 | .992 | .979 | .953 | .899 | .802 | .643 | .415 | .153 |
|     | 13  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .998 | .993 | .982 | .956 | .897 | .771 | .512 |
| 15  | 0   | .463 | .206 | .087 | .035 | .013 | .005 | .002 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 |
|     | 1   | .829 | .549 | .319 | .167 | .080 | .035 | .014 | .005 | .002 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 |
|     | 2   | .964 | .816 | .604 | .398 | .236 | .127 | .062 | .027 | .011 | .004 | .001 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 |
|     | 3   | .995 | .944 | .823 | .648 | .461 | .297 | .173 | .091 | .042 | .018 | .006 | .002 | .000 | .000 | .000 | .000 | .000 | .000 | .000 |
|     | 4   | .999 | .987 | .938 | .836 | .686 | .515 | .352 | .217 | .120 | .059 | .025 | .009 | .003 | .001 | .000 | .000 | .000 | .000 | .000 |
|     | 5   | 1.0  | .998 | .983 | .939 | .852 | .722 | .564 | .403 | .261 | .151 | .077 | .034 | .012 | .004 | .001 | .000 | .000 | .000 | .000 |
|     | 6   | 1.0  | 1.0  | .996 | .982 | .943 | .869 | .755 | .610 | .452 | .304 | .182 | .095 | .042 | .015 | .004 | .001 | .000 | .000 | .000 |
|     | 7   | 1.0  | 1.0  | .999 | .996 | .983 | .950 | .887 | .787 | .654 | .500 | .346 | .213 | .113 | .050 | .017 | .004 | .001 | .000 | .000 |
|     | 8   | 1.0  | 1.0  | 1.0  | .999 | .996 | .985 | .958 | .905 | .818 | .696 | .548 | .390 | .245 | .131 | .057 | .018 | .004 | .000 | .000 |
|     | 9   | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .996 | .988 | .966 | .923 | .849 | .739 | .597 | .436 | .278 | .148 | .061 | .017 | .002 | .000 |
|     | 10  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .997 | .991 | .975 | .941 | .880 | .783 | .648 | .485 | .314 | .164 | .062 | .013 | .001 |
|     | 11  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .998 | .994 | .982 | .958 | .909 | .827 | .703 | .539 | .352 | .177 | .056 | .005 |
|     | 12  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .996 | .989 | .973 | .938 | .873 | .764 | .602 | .396 | .184 | .036 |
|     | 13  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .998 | .995 | .986 | .965 | .920 | .833 | .681 | .451 | .171 |
|     | 14  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .998 | .995 | .987 | .965 | .913 | .794 | .537 |

Table A2, continued. Binomial distribution

| n  | x   | p    |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
|----|-----|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
|    |     | .050 | .100 | .150 | .200 | .250 | .300 | .350 | .400 | .450 | .500 | .550 | .600 | .650 | .700 | .750 | .800 | .850 | .900 | .950 |
| 16 | 1   | .811 | .515 | .284 | .141 | .063 | .026 | .010 | .003 | .001 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 |
|    | 2   | .957 | .789 | .561 | .352 | .197 | .099 | .045 | .018 | .007 | .002 | .001 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 |
|    | 3   | .993 | .932 | .790 | .598 | .405 | .246 | .134 | .065 | .028 | .011 | .003 | .001 | .000 | .000 | .000 | .000 | .000 | .000 | .000 |
|    | 4   | .999 | .983 | .921 | .798 | .630 | .450 | .289 | .167 | .085 | .038 | .015 | .005 | .001 | .000 | .000 | .000 | .000 | .000 | .000 |
|    | 5   | 1.0  | .997 | .976 | .918 | .810 | .660 | .490 | .329 | .198 | .105 | .049 | .019 | .006 | .002 | .000 | .000 | .000 | .000 | .000 |
|    | 6   | 1.0  | .999 | .994 | .973 | .920 | .825 | .688 | .527 | .366 | .227 | .124 | .058 | .023 | .007 | .002 | .000 | .000 | .000 | .000 |
|    | 7   | 1.0  | 1.0  | .999 | .993 | .973 | .926 | .841 | .716 | .563 | .402 | .256 | .142 | .067 | .026 | .007 | .001 | .000 | .000 | .000 |
|    | 8   | 1.0  | 1.0  | 1.0  | .999 | .993 | .974 | .933 | .858 | .744 | .598 | .437 | .284 | .159 | .074 | .027 | .007 | .001 | .000 | .000 |
|    | 9   | 1.0  | 1.0  | 1.0  | 1.0  | .998 | .993 | .977 | .942 | .876 | .773 | .634 | .473 | .312 | .175 | .080 | .027 | .006 | .001 | .000 |
|    | 10  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .998 | .994 | .981 | .951 | .895 | .802 | .671 | .510 | .340 | .190 | .082 | .024 | .003 | .000 |
|    | 11  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .995 | .985 | .962 | .915 | .833 | .711 | .550 | .370 | .202 | .079 | .017 | .001 |
|    | 12  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .997 | .989 | .972 | .935 | .866 | .754 | .595 | .402 | .210 | .068 | .007 |
|    | 13  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .998 | .993 | .982 | .955 | .901 | .803 | .648 | .439 | .211 | .043 |
|    | 14  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .997 | .990 | .974 | .937 | .859 | .716 | .485 | .189 |
|    | 15  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .999 | .997 | .990 | .972 | .926 | .815 | .560 |
| 18 | 1   | .774 | .450 | .224 | .099 | .039 | .014 | .005 | .001 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 |
|    | 2   | .942 | .734 | .480 | .271 | .135 | .060 | .024 | .008 | .003 | .001 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 |
|    | 3   | .989 | .902 | .720 | .501 | .306 | .165 | .078 | .033 | .012 | .004 | .001 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 |
|    | y 4 | .998 | .972 | .879 | .716 | .519 | .333 | .189 | .094 | .041 | .015 | .005 | .001 | .000 | .000 | .000 | .000 | .000 | .000 | .000 |
|    | 5   | 1.0  | .994 | .958 | .867 | .717 | .534 | .355 | .209 | .108 | .048 | .018 | .006 | .001 | .000 | .000 | .000 | .000 | .000 | .000 |
|    | 6   | 1.0  | .999 | .988 | .949 | .861 | .722 | .549 | .374 | .226 | .119 | .054 | .020 | .006 | .001 | .000 | .000 | .000 | .000 | .000 |
|    | 7   | 1.0  | 1.0  | .997 | .984 | .943 | .859 | .728 | .563 | .391 | .240 | .128 | .058 | .021 | .006 | .001 | .000 | .000 | .000 | .000 |
|    | 8   | 1.0  | 1.0  | .999 | .996 | .981 | .940 | .861 | .737 | .578 | .407 | .253 | .135 | .060 | .021 | .005 | .001 | .000 | .000 | .000 |
|    | 9   | 1.0  | 1.0  | 1.0  | .999 | .995 | .979 | .940 | .865 | .747 | .593 | .422 | .263 | .139 | .060 | .019 | .004 | .001 | .000 | .000 |
|    | 10  | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .994 | .979 | .942 | .872 | .760 | .609 | .437 | .272 | .141 | .057 | .016 | .003 | .000 | .000 |
|    | 11  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .994 | .980 | .946 | .881 | .774 | .626 | .451 | .278 | .139 | .051 | .012 | .001 | .000 |
|    | 12  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .994 | .982 | .952 | .892 | .791 | .645 | .466 | .283 | .133 | .042 | .006 | .000 |
|    | 13  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .995 | .985 | .959 | .906 | .811 | .667 | .481 | .284 | .121 | .028 | .002 |
|    | 14  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .996 | .988 | .967 | .922 | .835 | .694 | .499 | .280 | .098 | .011 |
|    | 15  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .997 | .992 | .976 | .940 | .865 | .729 | .520 | .266 | .058 |
|    | 16  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .995 | .986 | .961 | .901 | .776 | .550 | .226 |
| 20 | 1   | .736 | .392 | .176 | .069 | .024 | .008 | .002 | .001 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 |
|    | 2   | .925 | .677 | .405 | .206 | .091 | .035 | .012 | .004 | .001 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 |
|    | y 3 | .984 | .867 | .648 | .411 | .225 | .107 | .044 | .016 | .005 | .001 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 |
|    | 4   | .997 | .957 | .830 | .630 | .415 | .238 | .118 | .051 | .019 | .006 | .002 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 |
|    | 5   | 1.0  | .989 | .933 | .804 | .617 | .416 | .245 | .126 | .055 | .021 | .006 | .002 | .000 | .000 | .000 | .000 | .000 | .000 | .000 |
|    | 6   | 1.0  | .998 | .978 | .913 | .786 | .608 | .417 | .250 | .130 | .058 | .021 | .006 | .002 | .000 | .000 | .000 | .000 | .000 | .000 |
|    | 7   | 1.0  | 1.0  | .994 | .968 | .898 | .772 | .601 | .416 | .252 | .132 | .058 | .021 | .006 | .001 | .000 | .000 | .000 | .000 | .000 |
|    | 8   | 1.0  | 1.0  | .999 | .990 | .959 | .887 | .762 | .596 | .414 | .252 | .131 | .057 | .020 | .005 | .001 | .000 | .000 | .000 | .000 |
|    | 9   | 1.0  | 1.0  | 1.0  | .997 | .986 | .952 | .878 | .755 | .591 | .412 | .249 | .128 | .053 | .017 | .004 | .001 | .000 | .000 | .000 |
|    | 10  | 1.0  | 1.0  | 1.0  | .999 | .996 | .983 | .947 | .872 | .751 | .588 | .409 | .245 | .122 | .048 | .014 | .003 | .000 | .000 | .000 |
|    | 11  | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .995 | .980 | .943 | .869 | .748 | .586 | .404 | .238 | .113 | .041 | .010 | .001 | .000 | .000 |
|    | 12  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .994 | .979 | .942 | .868 | .748 | .584 | .399 | .228 | .102 | .032 | .006 | .000 | .000 |
|    | 13  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .998 | .994 | .979 | .942 | .870 | .750 | .583 | .392 | .214 | .087 | .022 | .002 | .000 |
|    | 14  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .998 | .994 | .979 | .945 | .874 | .755 | .584 | .383 | .196 | .067 | .011 | .000 |
|    | 15  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .998 | .994 | .981 | .949 | .882 | .762 | .585 | .370 | .170 | .043 | .003 |
|    | 16  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .995 | .984 | .956 | .893 | .775 | .589 | .352 | .133 |
| 25 | 2   | .873 | .537 | .254 | .098 | .032 | .009 | .002 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 |
|    | 3   | .966 | .764 | .471 | .234 | .096 | .033 | .010 | .002 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 |
|    | 4   | .993 | .902 | .682 | .421 | .214 | .090 | .032 | .009 | .002 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 |
|    | 5   | .999 | .967 | .838 | .617 | .378 | .193 | .083 | .029 | .009 | .002 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 |
|    | 6   | 1.0  | .991 | .930 | .780 | .561 | .341 | .173 | .074 | .026 | .007 | .002 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 |
|    | 7   | 1.0  | .998 | .975 | .891 | .727 | .512 | .306 | .154 | .064 | .022 | .006 | .001 | .000 | .000 | .000 | .000 | .000 | .000 | .000 |
|    | 8   | 1.0  | 1.0  | .992 | .953 | .851 | .677 | .467 | .274 | .134 | .054 | .017 | .004 | .001 | .000 | .000 | .000 | .000 | .000 | .000 |
|    | 9   | 1.0  | 1.0  | .998 | .983 | .929 | .811 | .630 | .425 | .242 | .115 | .044 | .013 | .003 | .000 | .000 | .000 | .000 | .000 | .000 |
|    | 10  | 1.0  | 1.0  | 1.0  | .994 | .970 | .902 | .771 | .586 | .384 | .212 | .096 | .034 | .009 | .002 | .000 | .000 | .000 | .000 | .000 |
|    | 11  | 1.0  | 1.0  | 1.0  | .998 | .989 | .956 | .875 | .732 | .543 | .345 | .183 | .078 | .025 | .006 | .001 | .000 | .000 | .000 | .000 |
|    | 12  | 1.0  | 1.0  | 1.0  | 1.0  | .997 | .983 | .940 | .846 | .694 | .500 | .306 | .154 | .060 | .017 | .003 | .000 | .000 | .000 | .000 |
|    | 13  | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .994 | .975 | .922 | .817 | .655 | .457 | .268 | .125 | .044 | .011 | .002 | .000 | .000 | .000 |
|    | 14  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .998 | .991 | .966 | .904 | .788 | .616 | .414 | .229 | .098 | .030 | .006 | .000 | .000 | .000 |
|    | 15  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .997 | .987 | .956 | .885 | .758 | .575 | .370 | .189 | .071 | .017 | .002 | .000 | .000 |
|    | 16  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .996 | .983 | .946 | .866 | .726 | .533 | .323 | .149 | .047 | .008 | .000 | .000 |
|    | 17  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .999 | .994 | .978 | .936 | .846 | .694 | .488 | .273 | .109 | .025 | .002 | .000 |
|    | 18  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .998 | .993 | .974 | .926 | .827 | .659 | .439 | .220 | .070 | .009 | .000 |
|    | 19  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .998 | .991 | .971 | .917 | .807 | .622 | .383 | .162 | .033 | .001 |
|    | 20  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | .998 | .991 | .968 | .910 | .786 | .579 | .318 | .098 | .007 |

**Table A3. Poisson distribution**

$$F(x) = P\{X \leq x\} = \sum_{k=0}^x \frac{e^{-\lambda} \lambda^k}{k!}$$

| $x$ | $\lambda$ |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
|-----|-----------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
|     | 0.1       | 0.2  | 0.3  | 0.4  | 0.5  | 0.6  | 0.7  | 0.8  | 0.9  | 1.0  | 1.1  | 1.2  | 1.3  | 1.4  | 1.5  |
| 0   | .905      | .819 | .741 | .670 | .607 | .549 | .497 | .449 | .407 | .368 | .333 | .301 | .273 | .247 | .223 |
| 1   | .995      | .982 | .963 | .938 | .910 | .878 | .844 | .809 | .772 | .736 | .699 | .663 | .627 | .592 | .558 |
| 2   | 1.00      | .999 | .996 | .992 | .986 | .977 | .966 | .953 | .937 | .920 | .900 | .879 | .857 | .833 | .809 |
| 3   | 1.00      | 1.00 | 1.00 | .999 | .998 | .997 | .994 | .991 | .987 | .981 | .974 | .966 | .957 | .946 | .934 |
| 4   | 1.00      | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | .999 | .999 | .998 | .996 | .995 | .992 | .989 | .986 | .981 |
| 5   | 1.00      | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | .999 | .999 | .998 | .998 | .997 | .996 |
| 6   | 1.00      | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | .999 | .999 |
| 7   | 1.00      | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |

| $x$ | $\lambda$ |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
|-----|-----------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
|     | 1.6       | 1.7  | 1.8  | 1.9  | 2.0  | 2.1  | 2.2  | 2.3  | 2.4  | 2.5  | 2.6  | 2.7  | 2.8  | 2.9  | 3.0  |
| 0   | .202      | .183 | .165 | .150 | .135 | .122 | .111 | .100 | .091 | .082 | .074 | .067 | .061 | .055 | .050 |
| 1   | .525      | .493 | .463 | .434 | .406 | .380 | .355 | .331 | .308 | .287 | .267 | .249 | .231 | .215 | .199 |
| 2   | .783      | .757 | .731 | .704 | .677 | .650 | .623 | .596 | .570 | .544 | .518 | .494 | .469 | .446 | .423 |
| 3   | .921      | .907 | .891 | .875 | .857 | .839 | .819 | .799 | .779 | .758 | .736 | .714 | .692 | .670 | .647 |
| 4   | .976      | .970 | .964 | .956 | .947 | .938 | .928 | .916 | .904 | .891 | .877 | .863 | .848 | .832 | .815 |
| 5   | .994      | .992 | .990 | .987 | .983 | .980 | .975 | .970 | .964 | .958 | .951 | .943 | .935 | .926 | .916 |
| 6   | .999      | .998 | .997 | .997 | .995 | .994 | .993 | .991 | .988 | .986 | .983 | .979 | .976 | .971 | .966 |
| 7   | 1.00      | 1.00 | .999 | .999 | .999 | .999 | .998 | .997 | .997 | .996 | .995 | .993 | .992 | .990 | .988 |
| 8   | 1.00      | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | .999 | .999 | .999 | .999 | .998 | .998 | .997 | .996 |
| 9   | 1.00      | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | .999 | .999 | .999 | .999 |
| 10  | 1.00      | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |

| $x$ | $\lambda$ |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
|-----|-----------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
|     | 3.5       | 4.0  | 4.5  | 5.0  | 5.5  | 6.0  | 6.5  | 7.0  | 7.5  | 8.0  | 8.5  | 9.0  | 9.5  | 10.0 | 10.5 |
| 0   | .030      | .018 | .011 | .007 | .004 | .002 | .002 | .001 | .001 | .000 | .000 | .000 | .000 | .000 | .000 |
| 1   | .136      | .092 | .061 | .040 | .027 | .017 | .011 | .007 | .005 | .003 | .002 | .001 | .001 | .000 | .000 |
| 2   | .321      | .238 | .174 | .125 | .088 | .062 | .043 | .030 | .020 | .014 | .009 | .006 | .004 | .003 | .002 |
| 3   | .537      | .433 | .342 | .265 | .202 | .151 | .112 | .082 | .059 | .042 | .030 | .021 | .015 | .010 | .007 |
| 4   | .725      | .629 | .532 | .440 | .358 | .285 | .224 | .173 | .132 | .100 | .074 | .055 | .040 | .029 | .021 |
| 5   | .858      | .785 | .703 | .616 | .529 | .446 | .369 | .301 | .241 | .191 | .150 | .116 | .089 | .067 | .050 |
| 6   | .935      | .889 | .831 | .762 | .686 | .606 | .527 | .450 | .378 | .313 | .256 | .207 | .165 | .130 | .102 |
| 7   | .973      | .949 | .913 | .867 | .809 | .744 | .673 | .599 | .525 | .453 | .386 | .324 | .269 | .220 | .179 |
| 8   | .990      | .979 | .960 | .932 | .894 | .847 | .792 | .729 | .662 | .593 | .523 | .456 | .392 | .333 | .279 |
| 9   | .997      | .992 | .983 | .968 | .946 | .916 | .877 | .830 | .776 | .717 | .653 | .587 | .522 | .458 | .397 |
| 10  | .999      | .997 | .993 | .986 | .975 | .957 | .933 | .901 | .862 | .816 | .763 | .706 | .645 | .583 | .521 |
| 11  | 1.00      | .999 | .998 | .995 | .989 | .980 | .966 | .947 | .921 | .888 | .849 | .803 | .752 | .697 | .639 |
| 12  | 1.00      | 1.00 | .999 | .998 | .996 | .991 | .984 | .973 | .957 | .936 | .909 | .876 | .836 | .792 | .742 |
| 13  | 1.00      | 1.00 | 1.00 | .999 | .998 | .996 | .993 | .987 | .978 | .966 | .949 | .926 | .898 | .864 | .825 |
| 14  | 1.00      | 1.00 | 1.00 | 1.00 | .999 | .999 | .997 | .994 | .990 | .983 | .973 | .959 | .940 | .917 | .888 |
| 15  | 1.00      | 1.00 | 1.00 | 1.00 | 1.00 | .999 | .999 | .998 | .995 | .992 | .986 | .978 | .967 | .951 | .932 |
| 16  | 1.00      | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | .999 | .998 | .996 | .993 | .989 | .982 | .973 | .960 |
| 17  | 1.00      | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | .999 | .998 | .997 | .995 | .991 | .986 | .978 |
| 18  | 1.00      | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | .999 | .999 | .998 | .996 | .993 | .988 |
| 19  | 1.00      | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | .999 | .999 | .998 | .997 | .994 |
| 20  | 1.00      | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | .999 | .998 | .997 |

Table A3, continued. Poisson distribution

| x  | $\lambda$ |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
|----|-----------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
|    | 11        | 12   | 13   | 14   | 15   | 16   | 17   | 18   | 19   | 20   | 22   | 24   | 26   | 28   | 30   |
| 0  | .000      | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 |
| 1  | .000      | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 |
| 2  | .001      | .001 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 |
| 3  | .005      | .002 | .001 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 |
| 4  | .015      | .008 | .004 | .002 | .001 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 |
| 5  | .038      | .020 | .011 | .006 | .003 | .001 | .001 | .000 | .000 | .000 | .000 | .000 | .000 | .000 | .000 |
| 6  | .079      | .046 | .026 | .014 | .008 | .004 | .002 | .001 | .001 | .000 | .000 | .000 | .000 | .000 | .000 |
| 7  | .143      | .090 | .054 | .032 | .018 | .010 | .005 | .003 | .002 | .001 | .000 | .000 | .000 | .000 | .000 |
| 8  | .232      | .155 | .100 | .062 | .037 | .022 | .013 | .007 | .004 | .002 | .001 | .000 | .000 | .000 | .000 |
| 9  | .341      | .242 | .166 | .109 | .070 | .043 | .026 | .015 | .009 | .005 | .002 | .000 | .000 | .000 | .000 |
| 10 | .460      | .347 | .252 | .176 | .118 | .077 | .049 | .030 | .018 | .011 | .004 | .001 | .000 | .000 | .000 |
| 11 | .579      | .462 | .353 | .260 | .185 | .127 | .085 | .055 | .035 | .021 | .008 | .003 | .001 | .000 | .000 |
| 12 | .689      | .576 | .463 | .358 | .268 | .193 | .135 | .092 | .061 | .039 | .015 | .005 | .002 | .001 | .000 |
| 13 | .781      | .682 | .573 | .464 | .363 | .275 | .201 | .143 | .098 | .066 | .028 | .011 | .004 | .001 | .000 |
| 14 | .854      | .772 | .675 | .570 | .466 | .368 | .281 | .208 | .150 | .105 | .048 | .020 | .008 | .003 | .001 |
| 15 | .907      | .844 | .764 | .669 | .568 | .467 | .371 | .287 | .215 | .157 | .077 | .034 | .014 | .005 | .002 |
| 16 | .944      | .899 | .835 | .756 | .664 | .566 | .468 | .375 | .292 | .221 | .117 | .056 | .025 | .010 | .004 |
| 17 | .968      | .937 | .890 | .827 | .749 | .659 | .564 | .469 | .378 | .297 | .169 | .087 | .041 | .018 | .007 |
| 18 | .982      | .963 | .930 | .883 | .819 | .742 | .655 | .562 | .469 | .381 | .232 | .128 | .065 | .030 | .013 |
| 19 | .991      | .979 | .957 | .923 | .875 | .812 | .736 | .651 | .561 | .470 | .306 | .180 | .097 | .048 | .022 |
| 20 | .995      | .988 | .975 | .952 | .917 | .868 | .805 | .731 | .647 | .559 | .387 | .243 | .139 | .073 | .035 |
| 21 | .998      | .994 | .986 | .971 | .947 | .911 | .861 | .799 | .725 | .644 | .472 | .314 | .190 | .106 | .054 |
| 22 | .999      | .997 | .992 | .983 | .967 | .942 | .905 | .855 | .793 | .721 | .556 | .392 | .252 | .148 | .081 |
| 23 | 1.00      | .999 | .996 | .991 | .981 | .963 | .937 | .899 | .849 | .787 | .637 | .473 | .321 | .200 | .115 |
| 24 | 1.00      | .999 | .998 | .995 | .989 | .978 | .959 | .932 | .893 | .843 | .712 | .554 | .396 | .260 | .157 |
| 25 | 1.00      | 1.00 | .999 | .997 | .994 | .987 | .975 | .955 | .927 | .888 | .777 | .632 | .474 | .327 | .208 |
| 26 | 1.00      | 1.00 | 1.00 | .999 | .997 | .993 | .985 | .972 | .951 | .922 | .832 | .704 | .552 | .400 | .267 |
| 27 | 1.00      | 1.00 | 1.00 | .999 | .998 | .996 | .991 | .983 | .969 | .948 | .877 | .768 | .627 | .475 | .333 |
| 28 | 1.00      | 1.00 | 1.00 | 1.00 | .999 | .998 | .995 | .990 | .980 | .966 | .913 | .823 | .697 | .550 | .403 |
| 29 | 1.00      | 1.00 | 1.00 | 1.00 | 1.00 | .999 | .997 | .994 | .988 | .978 | .940 | .868 | .759 | .623 | .476 |
| 30 | 1.00      | 1.00 | 1.00 | 1.00 | 1.00 | .999 | .999 | .997 | .993 | .987 | .959 | .904 | .813 | .690 | .548 |
| 31 | 1.00      | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | .999 | .998 | .996 | .992 | .973 | .932 | .859 | .752 | .619 |
| 32 | 1.00      | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | .999 | .998 | .995 | .983 | .953 | .896 | .805 | .685 |
| 33 | 1.00      | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | .999 | .997 | .989 | .969 | .925 | .850 | .744 |
| 34 | 1.00      | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | .999 | .999 | .994 | .979 | .947 | .888 | .797 |
| 35 | 1.00      | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | .999 | .996 | .987 | .964 | .918 | .843 |
| 36 | 1.00      | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | .998 | .992 | .976 | .941 | .880 |
| 37 | 1.00      | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | .999 | .995 | .984 | .959 | .911 |
| 38 | 1.00      | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | .999 | .997 | .990 | .972 | .935 |
| 39 | 1.00      | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | .998 | .994 | .981 | .954 |
| 40 | 1.00      | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | .999 | .996 | .988 | .968 |
| 41 | 1.00      | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | .999 | .998 | .992 | .978 |
| 42 | 1.00      | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | .999 | .995 | .985 |
| 43 | 1.00      | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | .999 | .997 | .990 |
| 44 | 1.00      | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | .998 | .994 |
| 45 | 1.00      | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | .999 | .996 |
| 46 | 1.00      | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | .999 | .998 |
| 47 | 1.00      | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | .999 |
| 48 | 1.00      | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | .999 |
| 49 | 1.00      | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | .999 |
| 50 | 1.00      | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |

Table A4. Standard Normal distribution

$$\Phi(z) = P\{Z \leq z\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-x^2/2} dx$$

| $z$     | -0.09 | -0.08 | -0.07 | -0.06 | -0.05 | -0.04 | -0.03 | -0.02 | -0.01 | -0.00 |
|---------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| -(3.9+) | .0000 | .0000 | .0000 | .0000 | .0000 | .0000 | .0000 | .0000 | .0000 | .0000 |
| -3.8    | .0001 | .0001 | .0001 | .0001 | .0001 | .0001 | .0001 | .0001 | .0001 | .0001 |
| -3.7    | .0001 | .0001 | .0001 | .0001 | .0001 | .0001 | .0001 | .0001 | .0001 | .0001 |
| -3.6    | .0001 | .0001 | .0001 | .0001 | .0001 | .0001 | .0001 | .0001 | .0002 | .0002 |
| -3.5    | .0002 | .0002 | .0002 | .0002 | .0002 | .0002 | .0002 | .0002 | .0002 | .0002 |
| -3.4    | .0002 | .0003 | .0003 | .0003 | .0003 | .0003 | .0003 | .0003 | .0003 | .0003 |
| -3.3    | .0003 | .0004 | .0004 | .0004 | .0004 | .0004 | .0004 | .0005 | .0005 | .0005 |
| -3.2    | .0005 | .0005 | .0005 | .0006 | .0006 | .0006 | .0006 | .0006 | .0007 | .0007 |
| -3.1    | .0007 | .0007 | .0008 | .0008 | .0008 | .0008 | .0009 | .0009 | .0009 | .0010 |
| -3.0    | .0010 | .0010 | .0011 | .0011 | .0011 | .0012 | .0012 | .0013 | .0013 | .0013 |
| -2.9    | .0014 | .0014 | .0015 | .0015 | .0016 | .0016 | .0017 | .0018 | .0018 | .0019 |
| -2.8    | .0019 | .0020 | .0021 | .0021 | .0022 | .0023 | .0023 | .0024 | .0025 | .0026 |
| -2.7    | .0026 | .0027 | .0028 | .0029 | .0030 | .0031 | .0032 | .0033 | .0034 | .0035 |
| -2.6    | .0036 | .0037 | .0038 | .0039 | .0040 | .0041 | .0043 | .0044 | .0045 | .0047 |
| -2.5    | .0048 | .0049 | .0051 | .0052 | .0054 | .0055 | .0057 | .0059 | .0060 | .0062 |
| -2.4    | .0064 | .0066 | .0068 | .0069 | .0071 | .0073 | .0075 | .0078 | .0080 | .0082 |
| -2.3    | .0084 | .0087 | .0089 | .0091 | .0094 | .0096 | .0099 | .0102 | .0104 | .0107 |
| -2.2    | .0110 | .0113 | .0116 | .0119 | .0122 | .0125 | .0129 | .0132 | .0136 | .0139 |
| -2.1    | .0143 | .0146 | .0150 | .0154 | .0158 | .0162 | .0166 | .0170 | .0174 | .0179 |
| -2.0    | .0183 | .0188 | .0192 | .0197 | .0202 | .0207 | .0212 | .0217 | .0222 | .0228 |
| -1.9    | .0233 | .0239 | .0244 | .0250 | .0256 | .0262 | .0268 | .0274 | .0281 | .0287 |
| -1.8    | .0294 | .0301 | .0307 | .0314 | .0322 | .0329 | .0336 | .0344 | .0351 | .0359 |
| -1.7    | .0367 | .0375 | .0384 | .0392 | .0401 | .0409 | .0418 | .0427 | .0436 | .0446 |
| -1.6    | .0455 | .0465 | .0475 | .0485 | .0495 | .0505 | .0516 | .0526 | .0537 | .0548 |
| -1.5    | .0559 | .0571 | .0582 | .0594 | .0606 | .0618 | .0630 | .0643 | .0655 | .0668 |
| -1.4    | .0681 | .0694 | .0708 | .0721 | .0735 | .0749 | .0764 | .0778 | .0793 | .0808 |
| -1.3    | .0823 | .0838 | .0853 | .0869 | .0885 | .0901 | .0918 | .0934 | .0951 | .0968 |
| -1.2    | .0985 | .1003 | .1020 | .1038 | .1056 | .1075 | .1093 | .1112 | .1131 | .1151 |
| -1.1    | .1170 | .1190 | .1210 | .1230 | .1251 | .1271 | .1292 | .1314 | .1335 | .1357 |
| -1.0    | .1379 | .1401 | .1423 | .1446 | .1469 | .1492 | .1515 | .1539 | .1562 | .1587 |
| -0.9    | .1611 | .1635 | .1660 | .1685 | .1711 | .1736 | .1762 | .1788 | .1814 | .1841 |
| -0.8    | .1867 | .1894 | .1922 | .1949 | .1977 | .2005 | .2033 | .2061 | .2090 | .2119 |
| -0.7    | .2148 | .2177 | .2206 | .2236 | .2266 | .2296 | .2327 | .2358 | .2389 | .2420 |
| -0.6    | .2451 | .2483 | .2514 | .2546 | .2578 | .2611 | .2643 | .2676 | .2709 | .2743 |
| -0.5    | .2776 | .2810 | .2843 | .2877 | .2912 | .2946 | .2981 | .3015 | .3050 | .3085 |
| -0.4    | .3121 | .3156 | .3192 | .3228 | .3264 | .3300 | .3336 | .3372 | .3409 | .3446 |
| -0.3    | .3483 | .3520 | .3557 | .3594 | .3632 | .3669 | .3707 | .3745 | .3783 | .3821 |
| -0.2    | .3859 | .3897 | .3936 | .3974 | .4013 | .4052 | .4090 | .4129 | .4168 | .4207 |
| -0.1    | .4247 | .4286 | .4325 | .4364 | .4404 | .4443 | .4483 | .4522 | .4562 | .4602 |
| -0.0    | .4641 | .4681 | .4721 | .4761 | .4801 | .4840 | .4880 | .4920 | .4960 | .5000 |

Table A4, continued. Standard Normal distribution

[illegible]



**Table A5. Table of Student's T-distribution** $t_\alpha$ ; critical values, such that  $P\{t > t_\alpha\} = \alpha$ 

| $\nu$<br>(d.f.) | $\alpha$ , the right-tail probability |       |        |       |       |       |       |       |       |       |
|-----------------|---------------------------------------|-------|--------|-------|-------|-------|-------|-------|-------|-------|
|                 | .10                                   | .05   | .025   | .02   | .01   | .005  | .0025 | .001  | .0005 | .0001 |
| 1               | 3.078                                 | 6.314 | 12.706 | 15.89 | 31.82 | 63.66 | 127.3 | 318.3 | 636.6 | 3185  |
| 2               | 1.886                                 | 2.920 | 4.303  | 4.849 | 6.965 | 9.925 | 14.09 | 22.33 | 31.60 | 70.71 |
| 3               | 1.638                                 | 2.353 | 3.182  | 3.482 | 4.541 | 5.841 | 7.453 | 10.21 | 12.92 | 22.20 |
| 4               | 1.533                                 | 2.132 | 2.776  | 2.999 | 3.747 | 4.604 | 5.598 | 7.173 | 8.610 | 13.04 |
| 5               | 1.476                                 | 2.015 | 2.571  | 2.757 | 3.365 | 4.032 | 4.773 | 5.894 | 6.869 | 9.676 |
| 6               | 1.440                                 | 1.943 | 2.447  | 2.612 | 3.143 | 3.707 | 4.317 | 5.208 | 5.959 | 8.023 |
| 7               | 1.415                                 | 1.895 | 2.365  | 2.517 | 2.998 | 3.499 | 4.029 | 4.785 | 5.408 | 7.064 |
| 8               | 1.397                                 | 1.860 | 2.306  | 2.449 | 2.896 | 3.355 | 3.833 | 4.501 | 5.041 | 6.442 |
| 9               | 1.383                                 | 1.833 | 2.262  | 2.398 | 2.821 | 3.250 | 3.690 | 4.297 | 4.781 | 6.009 |
| 10              | 1.372                                 | 1.812 | 2.228  | 2.359 | 2.764 | 3.169 | 3.581 | 4.144 | 4.587 | 5.694 |
| 11              | 1.363                                 | 1.796 | 2.201  | 2.328 | 2.718 | 3.106 | 3.497 | 4.025 | 4.437 | 5.453 |
| 12              | 1.356                                 | 1.782 | 2.179  | 2.303 | 2.681 | 3.055 | 3.428 | 3.930 | 4.318 | 5.263 |
| 13              | 1.350                                 | 1.771 | 2.160  | 2.282 | 2.650 | 3.012 | 3.372 | 3.852 | 4.221 | 5.111 |
| 14              | 1.345                                 | 1.761 | 2.145  | 2.264 | 2.624 | 2.977 | 3.326 | 3.787 | 4.140 | 4.985 |
| 15              | 1.341                                 | 1.753 | 2.131  | 2.249 | 2.602 | 2.947 | 3.286 | 3.733 | 4.073 | 4.880 |
| 16              | 1.337                                 | 1.746 | 2.120  | 2.235 | 2.583 | 2.921 | 3.252 | 3.686 | 4.015 | 4.790 |
| 17              | 1.333                                 | 1.740 | 2.110  | 2.224 | 2.567 | 2.898 | 3.222 | 3.646 | 3.965 | 4.715 |
| 18              | 1.330                                 | 1.734 | 2.101  | 2.214 | 2.552 | 2.878 | 3.197 | 3.610 | 3.922 | 4.648 |
| 19              | 1.328                                 | 1.729 | 2.093  | 2.205 | 2.539 | 2.861 | 3.174 | 3.579 | 3.883 | 4.590 |
| 20              | 1.325                                 | 1.725 | 2.086  | 2.197 | 2.528 | 2.845 | 3.153 | 3.552 | 3.850 | 4.539 |
| 21              | 1.323                                 | 1.721 | 2.080  | 2.189 | 2.518 | 2.831 | 3.135 | 3.527 | 3.819 | 4.492 |
| 22              | 1.321                                 | 1.717 | 2.074  | 2.183 | 2.508 | 2.819 | 3.119 | 3.505 | 3.792 | 4.452 |
| 23              | 1.319                                 | 1.714 | 2.069  | 2.177 | 2.500 | 2.807 | 3.104 | 3.485 | 3.768 | 4.416 |
| 24              | 1.318                                 | 1.711 | 2.064  | 2.172 | 2.492 | 2.797 | 3.091 | 3.467 | 3.745 | 4.382 |
| 25              | 1.316                                 | 1.708 | 2.060  | 2.167 | 2.485 | 2.787 | 3.078 | 3.450 | 3.725 | 4.352 |
| 26              | 1.315                                 | 1.706 | 2.056  | 2.162 | 2.479 | 2.779 | 3.067 | 3.435 | 3.707 | 4.324 |
| 27              | 1.314                                 | 1.703 | 2.052  | 2.158 | 2.473 | 2.771 | 3.057 | 3.421 | 3.689 | 4.299 |
| 28              | 1.313                                 | 1.701 | 2.048  | 2.154 | 2.467 | 2.763 | 3.047 | 3.408 | 3.674 | 4.276 |
| 29              | 1.311                                 | 1.699 | 2.045  | 2.150 | 2.462 | 2.756 | 3.038 | 3.396 | 3.660 | 4.254 |
| 30              | 1.310                                 | 1.697 | 2.042  | 2.147 | 2.457 | 2.750 | 3.030 | 3.385 | 3.646 | 4.234 |
| 32              | 1.309                                 | 1.694 | 2.037  | 2.141 | 2.449 | 2.738 | 3.015 | 3.365 | 3.622 | 4.198 |
| 34              | 1.307                                 | 1.691 | 2.032  | 2.136 | 2.441 | 2.728 | 3.002 | 3.348 | 3.601 | 4.168 |
| 36              | 1.306                                 | 1.688 | 2.028  | 2.131 | 2.434 | 2.719 | 2.990 | 3.333 | 3.582 | 4.140 |
| 38              | 1.304                                 | 1.686 | 2.024  | 2.127 | 2.429 | 2.712 | 2.980 | 3.319 | 3.566 | 4.115 |
| 40              | 1.303                                 | 1.684 | 2.021  | 2.123 | 2.423 | 2.704 | 2.971 | 3.307 | 3.551 | 4.094 |
| 45              | 1.301                                 | 1.679 | 2.014  | 2.115 | 2.412 | 2.690 | 2.952 | 3.281 | 3.520 | 4.049 |
| 50              | 1.299                                 | 1.676 | 2.009  | 2.109 | 2.403 | 2.678 | 2.937 | 3.261 | 3.496 | 4.014 |
| 55              | 1.297                                 | 1.673 | 2.004  | 2.104 | 2.396 | 2.668 | 2.925 | 3.245 | 3.476 | 3.985 |
| 60              | 1.296                                 | 1.671 | 2.000  | 2.099 | 2.390 | 2.660 | 2.915 | 3.232 | 3.460 | 3.962 |
| 70              | 1.294                                 | 1.667 | 1.994  | 2.093 | 2.381 | 2.648 | 2.899 | 3.211 | 3.435 | 3.926 |
| 80              | 1.292                                 | 1.664 | 1.990  | 2.088 | 2.374 | 2.639 | 2.887 | 3.195 | 3.416 | 3.899 |
| 90              | 1.291                                 | 1.662 | 1.987  | 2.084 | 2.368 | 2.632 | 2.878 | 3.183 | 3.402 | 3.878 |
| 100             | 1.290                                 | 1.660 | 1.984  | 2.081 | 2.364 | 2.626 | 2.871 | 3.174 | 3.390 | 3.861 |
| 200             | 1.286                                 | 1.653 | 1.972  | 2.067 | 2.345 | 2.601 | 2.838 | 3.131 | 3.340 | 3.789 |
| $\infty$        | 1.282                                 | 1.645 | 1.960  | 2.054 | 2.326 | 2.576 | 2.807 | 3.090 | 3.290 | 3.719 |

**Table A6. Table of Chi-Square Distribution** $\chi^2_\alpha$ ; critical values, such that  $P\{\chi^2 > \chi^2_\alpha\} = \alpha$ 

| $\nu$<br>(d.f.) | $\alpha$ , the right-tail probability |      |      |      |      |      |      |      |      |      |      |      |      |      |
|-----------------|---------------------------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|
|                 | .999                                  | .995 | .99  | .975 | .95  | .90  | .80  | .20  | .10  | .05  | .025 | .01  | .005 | .001 |
| 1               | 0.00                                  | 0.00 | 0.00 | 0.00 | 0.00 | 0.02 | 0.06 | 1.64 | 2.71 | 3.84 | 5.02 | 6.63 | 7.88 | 10.8 |
| 2               | 0.00                                  | 0.01 | 0.02 | 0.05 | 0.10 | 0.21 | 0.45 | 3.22 | 4.61 | 5.99 | 7.38 | 9.21 | 10.6 | 13.8 |
| 3               | 0.02                                  | 0.07 | 0.11 | 0.22 | 0.35 | 0.58 | 1.01 | 4.64 | 6.25 | 7.81 | 9.35 | 11.3 | 12.8 | 16.3 |
| 4               | 0.09                                  | 0.21 | 0.30 | 0.48 | 0.71 | 1.06 | 1.65 | 5.99 | 7.78 | 9.49 | 11.1 | 13.3 | 14.9 | 18.5 |
| 5               | 0.21                                  | 0.41 | 0.55 | 0.83 | 1.15 | 1.61 | 2.34 | 7.29 | 9.24 | 11.1 | 12.8 | 15.1 | 16.7 | 20.5 |
| 6               | 0.38                                  | 0.68 | 0.87 | 1.24 | 1.64 | 2.20 | 3.07 | 8.56 | 10.6 | 12.6 | 14.4 | 16.8 | 18.5 | 22.5 |
| 7               | 0.60                                  | 0.99 | 1.24 | 1.69 | 2.17 | 2.83 | 3.82 | 9.80 | 12.0 | 14.1 | 16.0 | 18.5 | 20.3 | 24.3 |
| 8               | 0.86                                  | 1.34 | 1.65 | 2.18 | 2.73 | 3.49 | 4.59 | 11.0 | 13.4 | 15.5 | 17.5 | 20.1 | 22.0 | 26.1 |
| 9               | 1.15                                  | 1.73 | 2.09 | 2.70 | 3.33 | 4.17 | 5.38 | 12.2 | 14.7 | 16.9 | 19.0 | 21.7 | 23.6 | 27.9 |
| 10              | 1.48                                  | 2.16 | 2.56 | 3.25 | 3.94 | 4.87 | 6.18 | 13.4 | 16.0 | 18.3 | 20.5 | 23.2 | 25.2 | 29.6 |
| 11              | 1.83                                  | 2.60 | 3.05 | 3.82 | 4.57 | 5.58 | 6.99 | 14.6 | 17.3 | 19.7 | 21.9 | 24.7 | 26.8 | 31.3 |
| 12              | 2.21                                  | 3.07 | 3.57 | 4.40 | 5.23 | 6.30 | 7.81 | 15.8 | 18.5 | 21.0 | 23.3 | 26.2 | 28.3 | 32.9 |
| 13              | 2.62                                  | 3.57 | 4.11 | 5.01 | 5.89 | 7.04 | 8.63 | 17.0 | 19.8 | 22.4 | 24.7 | 27.7 | 29.8 | 34.5 |
| 14              | 3.04                                  | 4.07 | 4.66 | 5.63 | 6.57 | 7.79 | 9.47 | 18.2 | 21.1 | 23.7 | 26.1 | 29.1 | 31.3 | 36.1 |
| 15              | 3.48                                  | 4.60 | 5.23 | 6.26 | 7.26 | 8.55 | 10.3 | 19.3 | 22.3 | 25.0 | 27.5 | 30.6 | 32.8 | 37.7 |
| 16              | 3.94                                  | 5.14 | 5.81 | 6.91 | 7.96 | 9.31 | 11.1 | 20.5 | 23.5 | 26.3 | 28.8 | 32.0 | 34.3 | 39.3 |
| 17              | 4.42                                  | 5.70 | 6.41 | 7.56 | 8.67 | 10.1 | 12.0 | 21.6 | 24.8 | 27.6 | 30.2 | 33.4 | 35.7 | 40.8 |
| 18              | 4.90                                  | 6.26 | 7.01 | 8.23 | 9.39 | 10.9 | 12.9 | 22.8 | 26.0 | 28.9 | 31.5 | 34.8 | 37.2 | 42.3 |
| 19              | 5.41                                  | 6.84 | 7.63 | 8.91 | 10.1 | 11.7 | 13.7 | 23.9 | 27.2 | 30.1 | 32.9 | 36.2 | 38.6 | 43.8 |
| 20              | 5.92                                  | 7.43 | 8.26 | 9.59 | 10.9 | 12.4 | 14.6 | 25.0 | 28.4 | 31.4 | 34.2 | 37.6 | 40.0 | 45.3 |
| 21              | 6.45                                  | 8.03 | 8.90 | 10.3 | 11.6 | 13.2 | 15.4 | 26.2 | 29.6 | 32.7 | 35.5 | 38.9 | 41.4 | 46.8 |
| 22              | 6.98                                  | 8.64 | 9.54 | 11.0 | 12.3 | 14.0 | 16.3 | 27.3 | 30.8 | 33.9 | 36.8 | 40.3 | 42.8 | 48.3 |
| 23              | 7.53                                  | 9.26 | 10.2 | 11.7 | 13.1 | 14.8 | 17.2 | 28.4 | 32.0 | 35.2 | 38.1 | 41.6 | 44.2 | 49.7 |
| 24              | 8.08                                  | 9.89 | 10.9 | 12.4 | 13.8 | 15.7 | 18.1 | 29.6 | 33.2 | 36.4 | 39.4 | 43.0 | 45.6 | 51.2 |
| 25              | 8.65                                  | 10.5 | 11.5 | 13.1 | 14.6 | 16.5 | 18.9 | 30.7 | 34.4 | 37.7 | 40.6 | 44.3 | 46.9 | 52.6 |
| 26              | 9.22                                  | 11.2 | 12.2 | 13.8 | 15.4 | 17.3 | 19.8 | 31.8 | 35.6 | 38.9 | 41.9 | 45.6 | 48.3 | 54.1 |
| 27              | 9.80                                  | 11.8 | 12.9 | 14.6 | 16.2 | 18.1 | 20.7 | 32.9 | 36.7 | 40.1 | 43.2 | 47.0 | 49.6 | 55.5 |
| 28              | 10.4                                  | 12.5 | 13.6 | 15.3 | 16.9 | 18.9 | 21.6 | 34.0 | 37.9 | 41.3 | 44.5 | 48.3 | 51.0 | 56.9 |
| 29              | 11.0                                  | 13.1 | 14.3 | 16.0 | 17.7 | 19.8 | 22.5 | 35.1 | 39.1 | 42.6 | 45.7 | 49.6 | 52.3 | 58.3 |
| 30              | 11.6                                  | 13.8 | 15.0 | 16.8 | 18.5 | 20.6 | 23.4 | 36.3 | 40.3 | 43.8 | 47.0 | 50.9 | 53.7 | 59.7 |
| 31              | 12.2                                  | 14.5 | 15.7 | 17.5 | 19.3 | 21.4 | 24.3 | 37.4 | 41.4 | 45.0 | 48.2 | 52.2 | 55.0 | 61.1 |
| 32              | 12.8                                  | 15.1 | 16.4 | 18.3 | 20.1 | 22.3 | 25.1 | 38.5 | 42.6 | 46.2 | 49.5 | 53.5 | 56.3 | 62.5 |
| 33              | 13.4                                  | 15.8 | 17.1 | 19.0 | 20.9 | 23.1 | 26.0 | 39.6 | 43.7 | 47.4 | 50.7 | 54.8 | 57.6 | 63.9 |
| 34              | 14.1                                  | 16.5 | 17.8 | 19.8 | 21.7 | 24.0 | 26.9 | 40.7 | 44.9 | 48.6 | 52.0 | 56.1 | 59.0 | 65.2 |
| 35              | 14.7                                  | 17.2 | 18.5 | 20.6 | 22.5 | 24.8 | 27.8 | 41.8 | 46.1 | 49.8 | 53.2 | 57.3 | 60.3 | 66.6 |
| 36              | 15.3                                  | 17.9 | 19.2 | 21.3 | 23.3 | 25.6 | 28.7 | 42.9 | 47.2 | 51.0 | 54.4 | 58.6 | 61.6 | 68   |
| 37              | 16.0                                  | 18.6 | 20.0 | 22.1 | 24.1 | 26.5 | 29.6 | 44.0 | 48.4 | 52.2 | 55.7 | 59.9 | 62.9 | 69.3 |
| 38              | 16.6                                  | 19.3 | 20.7 | 22.9 | 24.9 | 27.3 | 30.5 | 45.1 | 49.5 | 53.4 | 56.9 | 61.2 | 64.2 | 70.7 |
| 39              | 17.3                                  | 20.0 | 21.4 | 23.7 | 25.7 | 28.2 | 31.4 | 46.2 | 50.7 | 54.6 | 58.1 | 62.4 | 65.5 | 72.1 |
| 40              | 17.9                                  | 20.7 | 22.2 | 24.4 | 26.5 | 29.1 | 32.3 | 47.3 | 51.8 | 55.8 | 59.3 | 63.7 | 66.8 | 73.4 |
| 41              | 18.6                                  | 21.4 | 22.9 | 25.2 | 27.3 | 29.9 | 33.3 | 48.4 | 52.9 | 56.9 | 60.6 | 65.0 | 68.1 | 74.7 |
| 42              | 19.2                                  | 22.1 | 23.7 | 26.0 | 28.1 | 30.8 | 34.2 | 49.5 | 54.1 | 58.1 | 61.8 | 66.2 | 69.3 | 76.1 |
| 43              | 19.9                                  | 22.9 | 24.4 | 26.8 | 29.0 | 31.6 | 35.1 | 50.5 | 55.2 | 59.3 | 63.0 | 67.5 | 70.6 | 77.4 |
| 44              | 20.6                                  | 23.6 | 25.1 | 27.6 | 29.8 | 32.5 | 36.0 | 51.6 | 56.4 | 60.5 | 64.2 | 68.7 | 71.9 | 78.7 |
| 45              | 21.3                                  | 24.3 | 25.9 | 28.4 | 30.6 | 33.4 | 36.9 | 52.7 | 57.5 | 61.7 | 65.4 | 70.0 | 73.2 | 80.1 |
| 46              | 21.9                                  | 25.0 | 26.7 | 29.2 | 31.4 | 34.2 | 37.8 | 53.8 | 58.6 | 62.8 | 66.6 | 71.2 | 74.4 | 81.4 |
| 47              | 22.6                                  | 25.8 | 27.4 | 30.0 | 32.3 | 35.1 | 38.7 | 54.9 | 59.8 | 64.0 | 67.8 | 72.4 | 75.7 | 82.7 |
| 48              | 23.3                                  | 26.5 | 28.2 | 30.8 | 33.1 | 35.9 | 39.6 | 56.0 | 60.9 | 65.2 | 69.0 | 73.7 | 77.0 | 84.0 |
| 49              | 24.0                                  | 27.2 | 28.9 | 31.6 | 33.9 | 36.8 | 40.5 | 57.1 | 62.0 | 66.3 | 70.2 | 74.9 | 78.2 | 85.4 |
| 50              | 24.7                                  | 28.0 | 29.7 | 32.4 | 34.8 | 37.7 | 41.4 | 58.2 | 63.2 | 67.5 | 71.4 | 76.2 | 79.5 | 86.7 |

**Table A7. Table of F-distribution** $F_\alpha$ ; critical values such that  $P\{F > F_\alpha\} = \alpha$ 

| $\nu_2$ ,<br>denom.<br>d.f. | $\alpha$ | $\nu_1$ , numerator degrees of freedom |        |        |        |        |        |        |        |        |        |
|-----------------------------|----------|----------------------------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
|                             |          | 1                                      | 2      | 3      | 4      | 5      | 6      | 7      | 8      | 9      | 10     |
| 1                           | 0.25     | 5.83                                   | 7.5    | 8.2    | 8.58   | 8.82   | 8.98   | 9.1    | 9.19   | 9.26   | 9.32   |
|                             | 0.1      | 39.9                                   | 49.5   | 53.6   | 55.8   | 57.2   | 58.2   | 58.9   | 59.4   | 59.9   | 60.2   |
|                             | 0.05     | 161                                    | 199    | 216    | 225    | 230    | 234    | 237    | 239    | 241    | 242    |
|                             | 0.025    | 648                                    | 799    | 864    | 900    | 922    | 937    | 948    | 957    | 963    | 969    |
|                             | 0.01     | 4052                                   | 4999   | 5403   | 5625   | 5764   | 5859   | 5928   | 5981   | 6022   | 6056   |
|                             | 0.005    | 16211                                  | 19999  | 21615  | 22500  | 23056  | 23437  | 23715  | 23925  | 24091  | 24224  |
|                             | 0.001    | 405284                                 | 499999 | 540379 | 562500 | 576405 | 585937 | 592873 | 598144 | 602284 | 605621 |
| 2                           | 0.25     | 2.57                                   | 3      | 3.15   | 3.23   | 3.28   | 3.31   | 3.34   | 3.35   | 3.37   | 3.38   |
|                             | 0.1      | 8.53                                   | 9      | 9.16   | 9.24   | 9.29   | 9.33   | 9.35   | 9.37   | 9.38   | 9.39   |
|                             | 0.05     | 18.5                                   | 19     | 19.2   | 19.2   | 19.3   | 19.3   | 19.4   | 19.4   | 19.4   | 19.4   |
|                             | 0.025    | 38.5                                   | 39     | 39.2   | 39.2   | 39.3   | 39.3   | 39.4   | 39.4   | 39.4   | 39.4   |
|                             | 0.01     | 98.5                                   | 99     | 99.2   | 99.2   | 99.3   | 99.3   | 99.4   | 99.4   | 99.4   | 99.4   |
|                             | 0.005    | 199                                    | 199    | 199    | 199    | 199    | 199    | 199    | 199    | 199    | 199    |
|                             | 0.001    | 999                                    | 999    | 999    | 999    | 999    | 999    | 999    | 999    | 999    | 999    |
| 3                           | 0.25     | 2.02                                   | 2.28   | 2.36   | 2.39   | 2.41   | 2.42   | 2.43   | 2.44   | 2.44   | 2.44   |
|                             | 0.1      | 5.54                                   | 5.46   | 5.39   | 5.34   | 5.31   | 5.28   | 5.27   | 5.25   | 5.24   | 5.23   |
|                             | 0.05     | 10.1                                   | 9.55   | 9.28   | 9.12   | 9.01   | 8.94   | 8.89   | 8.85   | 8.81   | 8.79   |
|                             | 0.025    | 17.4                                   | 16     | 15.4   | 15.1   | 14.9   | 14.7   | 14.6   | 14.5   | 14.5   | 14.4   |
|                             | 0.01     | 34.1                                   | 30.8   | 29.5   | 28.7   | 28.2   | 27.9   | 27.7   | 27.5   | 27.3   | 27.2   |
|                             | 0.005    | 55.6                                   | 49.8   | 47.5   | 46.2   | 45.4   | 44.8   | 44.4   | 44.1   | 43.9   | 43.7   |
|                             | 0.001    | 167                                    | 149    | 141    | 137    | 135    | 133    | 132    | 131    | 130    | 129    |
| 4                           | 0.25     | 1.81                                   | 2      | 2.05   | 2.06   | 2.07   | 2.08   | 2.08   | 2.08   | 2.08   | 2.08   |
|                             | 0.1      | 4.54                                   | 4.32   | 4.19   | 4.11   | 4.05   | 4.01   | 3.98   | 3.95   | 3.94   | 3.92   |
|                             | 0.05     | 7.71                                   | 6.94   | 6.59   | 6.39   | 6.26   | 6.16   | 6.09   | 6.04   | 6      | 5.96   |
|                             | 0.025    | 12.2                                   | 10.6   | 9.98   | 9.6    | 9.36   | 9.2    | 9.07   | 8.98   | 8.9    | 8.84   |
|                             | 0.01     | 21.2                                   | 18     | 16.7   | 16     | 15.5   | 15.2   | 15     | 14.8   | 14.7   | 14.5   |
|                             | 0.005    | 31.3                                   | 26.3   | 24.3   | 23.2   | 22.5   | 22     | 21.6   | 21.4   | 21.1   | 21     |
|                             | 0.001    | 74.1                                   | 61.2   | 56.2   | 53.4   | 51.7   | 50.5   | 49.7   | 49     | 48.5   | 48.1   |
| 5                           | 0.25     | 1.69                                   | 1.85   | 1.88   | 1.89   | 1.89   | 1.89   | 1.89   | 1.89   | 1.89   | 1.89   |
|                             | 0.1      | 4.06                                   | 3.78   | 3.62   | 3.52   | 3.45   | 3.4    | 3.37   | 3.34   | 3.32   | 3.3    |
|                             | 0.05     | 6.61                                   | 5.79   | 5.41   | 5.19   | 5.05   | 4.95   | 4.88   | 4.82   | 4.77   | 4.74   |
|                             | 0.025    | 10                                     | 8.43   | 7.76   | 7.39   | 7.15   | 6.98   | 6.85   | 6.76   | 6.68   | 6.62   |
|                             | 0.01     | 16.3                                   | 13.3   | 12.1   | 11.4   | 11     | 10.7   | 10.5   | 10.3   | 10.2   | 10.1   |
|                             | 0.005    | 22.8                                   | 18.3   | 16.5   | 15.6   | 14.9   | 14.5   | 14.2   | 14     | 13.8   | 13.6   |
|                             | 0.001    | 47.2                                   | 37.1   | 33.2   | 31.1   | 29.8   | 28.8   | 28.2   | 27.6   | 27.2   | 26.9   |
| 6                           | 0.25     | 1.62                                   | 1.76   | 1.78   | 1.79   | 1.79   | 1.78   | 1.78   | 1.78   | 1.77   | 1.77   |
|                             | 0.1      | 3.78                                   | 3.46   | 3.29   | 3.18   | 3.11   | 3.05   | 3.01   | 2.98   | 2.96   | 2.94   |
|                             | 0.05     | 5.99                                   | 5.14   | 4.76   | 4.53   | 4.39   | 4.28   | 4.21   | 4.15   | 4.1    | 4.06   |
|                             | 0.025    | 8.81                                   | 7.26   | 6.6    | 6.23   | 5.99   | 5.82   | 5.7    | 5.6    | 5.52   | 5.46   |
|                             | 0.01     | 13.7                                   | 10.9   | 9.78   | 9.15   | 8.75   | 8.47   | 8.26   | 8.1    | 7.98   | 7.87   |
|                             | 0.005    | 18.6                                   | 14.5   | 12.9   | 12     | 11.5   | 11.1   | 10.8   | 10.6   | 10.4   | 10.3   |
|                             | 0.001    | 35.5                                   | 27     | 23.7   | 21.9   | 20.8   | 20     | 19.5   | 19     | 18.7   | 18.4   |
| 8                           | 0.25     | 1.54                                   | 1.66   | 1.67   | 1.66   | 1.66   | 1.65   | 1.64   | 1.64   | 1.63   | 1.63   |
|                             | 0.1      | 3.46                                   | 3.11   | 2.92   | 2.81   | 2.73   | 2.67   | 2.62   | 2.59   | 2.56   | 2.54   |
|                             | 0.05     | 5.32                                   | 4.46   | 4.07   | 3.84   | 3.69   | 3.58   | 3.5    | 3.44   | 3.39   | 3.35   |
|                             | 0.025    | 7.57                                   | 6.06   | 5.42   | 5.05   | 4.82   | 4.65   | 4.53   | 4.43   | 4.36   | 4.3    |
|                             | 0.01     | 11.3                                   | 8.65   | 7.59   | 7.01   | 6.63   | 6.37   | 6.18   | 6.03   | 5.91   | 5.81   |
|                             | 0.005    | 14.7                                   | 11     | 9.6    | 8.81   | 8.3    | 7.95   | 7.69   | 7.5    | 7.34   | 7.21   |
|                             | 0.001    | 25.4                                   | 18.5   | 15.8   | 14.4   | 13.5   | 12.9   | 12.4   | 12     | 11.8   | 11.5   |
| 10                          | 0.25     | 1.49                                   | 1.6    | 1.6    | 1.59   | 1.59   | 1.58   | 1.57   | 1.56   | 1.56   | 1.55   |
|                             | 0.1      | 3.29                                   | 2.92   | 2.73   | 2.61   | 2.52   | 2.46   | 2.41   | 2.38   | 2.35   | 2.32   |
|                             | 0.05     | 4.96                                   | 4.1    | 3.71   | 3.48   | 3.33   | 3.22   | 3.14   | 3.07   | 3.02   | 2.98   |
|                             | 0.025    | 6.94                                   | 5.46   | 4.83   | 4.47   | 4.24   | 4.07   | 3.95   | 3.85   | 3.78   | 3.72   |
|                             | 0.01     | 10                                     | 7.56   | 6.55   | 5.99   | 5.64   | 5.39   | 5.2    | 5.06   | 4.94   | 4.85   |
|                             | 0.005    | 12.8                                   | 9.43   | 8.08   | 7.34   | 6.87   | 6.54   | 6.3    | 6.12   | 5.97   | 5.85   |
|                             | 0.001    | 21                                     | 14.9   | 12.6   | 11.3   | 10.5   | 9.93   | 9.52   | 9.2    | 8.96   | 8.75   |

Table A7, continued. F-distribution

| $\nu_2$ ,<br>denom.<br>d.f. | $\alpha$ | $\nu_1$ , numerator degrees of freedom |        |        |        |        |        |        |        |        |          |
|-----------------------------|----------|----------------------------------------|--------|--------|--------|--------|--------|--------|--------|--------|----------|
|                             |          | 15                                     | 20     | 25     | 30     | 40     | 50     | 100    | 200    | 500    | $\infty$ |
| 1                           | 0.25     | 9.49                                   | 9.58   | 9.63   | 9.67   | 9.71   | 9.74   | 9.8    | 9.82   | 9.84   | 9.85     |
|                             | 0.1      | 61.2                                   | 61.7   | 62.1   | 62.3   | 62.5   | 62.7   | 63     | 63.2   | 63.3   | 63.3     |
|                             | 0.05     | 246                                    | 248    | 249    | 250    | 251    | 252    | 253    | 254    | 254    | 254      |
|                             | 0.025    | 985                                    | 993    | 998    | 1001   | 1006   | 1008   | 1013   | 1016   | 1017   | 1018     |
|                             | 0.01     | 6157                                   | 6209   | 6240   | 6261   | 6287   | 6303   | 6334   | 6350   | 6360   | 6366     |
|                             | 0.005    | 24630                                  | 24836  | 24960  | 25044  | 25148  | 25211  | 25337  | 25401  | 25439  | 25464    |
| 2                           | 0.001    | 615764                                 | 620908 | 624017 | 626099 | 628712 | 630285 | 633444 | 635030 | 635983 | 636619   |
|                             | 0.25     | 3.41                                   | 3.43   | 3.44   | 3.44   | 3.45   | 3.46   | 3.47   | 3.47   | 3.47   | 3.48     |
|                             | 0.1      | 9.42                                   | 9.44   | 9.45   | 9.46   | 9.47   | 9.47   | 9.48   | 9.49   | 9.49   | 9.49     |
|                             | 0.05     | 19.4                                   | 19.4   | 19.5   | 19.5   | 19.5   | 19.5   | 19.5   | 19.5   | 19.5   | 19.5     |
|                             | 0.025    | 39.4                                   | 39.4   | 39.5   | 39.5   | 39.5   | 39.5   | 39.5   | 39.5   | 39.5   | 39.5     |
|                             | 0.01     | 99.4                                   | 99.4   | 99.5   | 99.5   | 99.5   | 99.5   | 99.5   | 99.5   | 99.5   | 99.5     |
| 3                           | 0.005    | 199                                    | 199    | 199    | 199    | 199    | 199    | 199    | 199    | 199    | 199      |
|                             | 0.001    | 999                                    | 999    | 999    | 999    | 999    | 999    | 999    | 999    | 999    | 999      |
|                             | 0.25     | 2.46                                   | 2.46   | 2.46   | 2.47   | 2.47   | 2.47   | 2.47   | 2.47   | 2.47   | 2.47     |
|                             | 0.1      | 5.2                                    | 5.18   | 5.17   | 5.17   | 5.16   | 5.15   | 5.14   | 5.14   | 5.14   | 5.13     |
|                             | 0.05     | 8.7                                    | 8.66   | 8.63   | 8.62   | 8.59   | 8.58   | 8.55   | 8.54   | 8.53   | 8.53     |
|                             | 0.025    | 14.3                                   | 14.2   | 14.1   | 14.1   | 14     | 14     | 14     | 13.9   | 13.9   | 13.9     |
| 4                           | 0.01     | 26.9                                   | 26.7   | 26.6   | 26.5   | 26.4   | 26.4   | 26.2   | 26.2   | 26.1   | 26.1     |
|                             | 0.005    | 43.1                                   | 42.8   | 42.6   | 42.5   | 42.3   | 42.2   | 42     | 41.9   | 41.9   | 41.8     |
|                             | 0.001    | 127                                    | 126    | 126    | 125    | 125    | 125    | 124    | 124    | 124    | 123      |
|                             | 0.25     | 2.08                                   | 2.08   | 2.08   | 2.08   | 2.08   | 2.08   | 2.08   | 2.08   | 2.08   | 2.08     |
|                             | 0.1      | 3.87                                   | 3.84   | 3.83   | 3.82   | 3.8    | 3.8    | 3.78   | 3.77   | 3.76   | 3.76     |
|                             | 0.05     | 5.86                                   | 5.8    | 5.77   | 5.75   | 5.72   | 5.7    | 5.66   | 5.65   | 5.64   | 5.63     |
| 5                           | 0.025    | 8.66                                   | 8.56   | 8.5    | 8.46   | 8.41   | 8.38   | 8.32   | 8.29   | 8.27   | 8.26     |
|                             | 0.01     | 14.2                                   | 14     | 13.9   | 13.8   | 13.7   | 13.7   | 13.6   | 13.5   | 13.5   | 13.5     |
|                             | 0.005    | 20.4                                   | 20.2   | 20     | 19.9   | 19.8   | 19.7   | 19.5   | 19.4   | 19.4   | 19.3     |
|                             | 0.001    | 46.8                                   | 46.1   | 45.7   | 45.4   | 45.1   | 44.9   | 44.5   | 44.3   | 44.1   | 44.1     |
|                             | 0.25     | 1.89                                   | 1.88   | 1.88   | 1.88   | 1.88   | 1.88   | 1.87   | 1.87   | 1.87   | 1.87     |
|                             | 0.1      | 3.24                                   | 3.21   | 3.19   | 3.17   | 3.16   | 3.15   | 3.13   | 3.12   | 3.11   | 3.1      |
| 6                           | 0.05     | 4.62                                   | 4.56   | 4.52   | 4.5    | 4.46   | 4.44   | 4.41   | 4.39   | 4.37   | 4.36     |
|                             | 0.025    | 6.43                                   | 6.33   | 6.27   | 6.23   | 6.18   | 6.14   | 6.08   | 6.05   | 6.03   | 6.02     |
|                             | 0.01     | 9.72                                   | 9.55   | 9.45   | 9.38   | 9.29   | 9.24   | 9.13   | 9.08   | 9.04   | 9.02     |
|                             | 0.005    | 13.1                                   | 12.9   | 12.8   | 12.7   | 12.5   | 12.5   | 12.3   | 12.2   | 12.2   | 12.1     |
|                             | 0.001    | 25.9                                   | 25.4   | 25.1   | 24.9   | 24.6   | 24.4   | 24.1   | 24     | 23.9   | 23.8     |
|                             | 0.25     | 1.76                                   | 1.76   | 1.75   | 1.75   | 1.75   | 1.75   | 1.74   | 1.74   | 1.74   | 1.74     |
| 8                           | 0.1      | 2.87                                   | 2.84   | 2.81   | 2.8    | 2.78   | 2.77   | 2.75   | 2.73   | 2.73   | 2.72     |
|                             | 0.05     | 3.94                                   | 3.87   | 3.83   | 3.81   | 3.77   | 3.75   | 3.71   | 3.69   | 3.68   | 3.67     |
|                             | 0.025    | 5.27                                   | 5.17   | 5.11   | 5.07   | 5.01   | 4.98   | 4.92   | 4.88   | 4.86   | 4.85     |
|                             | 0.01     | 7.56                                   | 7.4    | 7.3    | 7.23   | 7.14   | 7.09   | 6.99   | 6.93   | 6.9    | 6.88     |
|                             | 0.005    | 9.81                                   | 9.59   | 9.45   | 9.36   | 9.24   | 9.17   | 9.03   | 8.95   | 8.91   | 8.88     |
|                             | 0.001    | 17.6                                   | 17.1   | 16.9   | 16.7   | 16.4   | 16.3   | 16     | 15.9   | 15.8   | 15.7     |
| 10                          | 0.25     | 1.62                                   | 1.61   | 1.6    | 1.6    | 1.59   | 1.59   | 1.58   | 1.58   | 1.58   | 1.58     |
|                             | 0.1      | 2.46                                   | 2.42   | 2.4    | 2.38   | 2.36   | 2.35   | 2.32   | 2.31   | 2.3    | 2.29     |
|                             | 0.05     | 3.22                                   | 3.15   | 3.11   | 3.08   | 3.04   | 3.02   | 2.97   | 2.95   | 2.94   | 2.93     |
|                             | 0.025    | 4.1                                    | 4      | 3.94   | 3.89   | 3.84   | 3.81   | 3.74   | 3.7    | 3.68   | 3.67     |
|                             | 0.01     | 5.52                                   | 5.36   | 5.26   | 5.2    | 5.12   | 5.07   | 4.96   | 4.91   | 4.88   | 4.86     |
|                             | 0.005    | 6.81                                   | 6.61   | 6.48   | 6.4    | 6.29   | 6.22   | 6.09   | 6.02   | 5.98   | 5.95     |
|                             | 0.001    | 10.8                                   | 10.5   | 10.3   | 10.1   | 9.92   | 9.8    | 9.57   | 9.45   | 9.38   | 9.33     |
|                             | 0.25     | 1.53                                   | 1.52   | 1.52   | 1.51   | 1.51   | 1.5    | 1.49   | 1.49   | 1.49   | 1.48     |
|                             | 0.1      | 2.24                                   | 2.2    | 2.17   | 2.16   | 2.13   | 2.12   | 2.09   | 2.07   | 2.06   | 2.06     |
|                             | 0.05     | 2.85                                   | 2.77   | 2.73   | 2.7    | 2.66   | 2.64   | 2.59   | 2.56   | 2.55   | 2.54     |
|                             | 0.025    | 3.52                                   | 3.42   | 3.35   | 3.31   | 3.26   | 3.22   | 3.15   | 3.12   | 3.09   | 3.08     |
|                             | 0.01     | 4.56                                   | 4.41   | 4.31   | 4.25   | 4.17   | 4.12   | 4.01   | 3.96   | 3.93   | 3.91     |
|                             | 0.005    | 5.47                                   | 5.27   | 5.15   | 5.07   | 4.97   | 4.9    | 4.77   | 4.71   | 4.67   | 4.64     |
|                             | 0.001    | 8.13                                   | 7.8    | 7.6    | 7.47   | 7.3    | 7.19   | 6.98   | 6.87   | 6.81   | 6.76     |

Table A7, continued. F-distribution

| $\nu_2$ ,<br>denom.<br>d.f. | $\alpha$ | $\nu_1$ , numerator degrees of freedom |      |      |      |      |      |      |      |      |      |
|-----------------------------|----------|----------------------------------------|------|------|------|------|------|------|------|------|------|
|                             |          | 1                                      | 2    | 3    | 4    | 5    | 6    | 7    | 8    | 9    | 10   |
| 15                          | 0.25     | 1.43                                   | 1.52 | 1.52 | 1.51 | 1.49 | 1.48 | 1.47 | 1.46 | 1.46 | 1.45 |
|                             | 0.1      | 3.07                                   | 2.7  | 2.49 | 2.36 | 2.27 | 2.21 | 2.16 | 2.12 | 2.09 | 2.06 |
|                             | 0.05     | 4.54                                   | 3.68 | 3.29 | 3.06 | 2.9  | 2.79 | 2.71 | 2.64 | 2.59 | 2.54 |
|                             | 0.025    | 6.2                                    | 4.77 | 4.15 | 3.8  | 3.58 | 3.41 | 3.29 | 3.2  | 3.12 | 3.06 |
|                             | 0.01     | 8.68                                   | 6.36 | 5.42 | 4.89 | 4.56 | 4.32 | 4.14 | 4    | 3.89 | 3.8  |
|                             | 0.005    | 10.8                                   | 7.7  | 6.48 | 5.8  | 5.37 | 5.07 | 4.85 | 4.67 | 4.54 | 4.42 |
| 20                          | 0.001    | 16.6                                   | 11.3 | 9.34 | 8.25 | 7.57 | 7.09 | 6.74 | 6.47 | 6.26 | 6.08 |
|                             | 0.25     | 1.4                                    | 1.49 | 1.48 | 1.47 | 1.45 | 1.44 | 1.43 | 1.42 | 1.41 | 1.4  |
|                             | 0.1      | 2.97                                   | 2.59 | 2.38 | 2.25 | 2.16 | 2.09 | 2.04 | 2    | 1.96 | 1.94 |
|                             | 0.05     | 4.35                                   | 3.49 | 3.1  | 2.87 | 2.71 | 2.6  | 2.51 | 2.45 | 2.39 | 2.35 |
|                             | 0.025    | 5.87                                   | 4.46 | 3.86 | 3.51 | 3.29 | 3.13 | 3.01 | 2.91 | 2.84 | 2.77 |
|                             | 0.01     | 8.1                                    | 5.85 | 4.94 | 4.43 | 4.1  | 3.87 | 3.7  | 3.56 | 3.46 | 3.37 |
| 25                          | 0.005    | 9.94                                   | 6.99 | 5.82 | 5.17 | 4.76 | 4.47 | 4.26 | 4.09 | 3.96 | 3.85 |
|                             | 0.001    | 14.8                                   | 9.95 | 8.1  | 7.1  | 6.46 | 6.02 | 5.69 | 5.44 | 5.24 | 5.08 |
|                             | 0.25     | 1.39                                   | 1.47 | 1.46 | 1.44 | 1.42 | 1.41 | 1.4  | 1.39 | 1.38 | 1.37 |
|                             | 0.1      | 2.92                                   | 2.53 | 2.32 | 2.18 | 2.09 | 2.02 | 1.97 | 1.93 | 1.89 | 1.87 |
|                             | 0.05     | 4.24                                   | 3.39 | 2.99 | 2.76 | 2.6  | 2.49 | 2.4  | 2.34 | 2.28 | 2.24 |
|                             | 0.025    | 5.69                                   | 4.29 | 3.69 | 3.35 | 3.13 | 2.97 | 2.85 | 2.75 | 2.68 | 2.61 |
| 30                          | 0.01     | 7.77                                   | 5.57 | 4.68 | 4.18 | 3.85 | 3.63 | 3.46 | 3.32 | 3.22 | 3.13 |
|                             | 0.005    | 9.48                                   | 6.6  | 5.46 | 4.84 | 4.43 | 4.15 | 3.94 | 3.78 | 3.64 | 3.54 |
|                             | 0.001    | 13.9                                   | 9.22 | 7.45 | 6.49 | 5.89 | 5.46 | 5.15 | 4.91 | 4.71 | 4.56 |
|                             | 0.25     | 1.38                                   | 1.45 | 1.44 | 1.42 | 1.41 | 1.39 | 1.38 | 1.37 | 1.36 | 1.35 |
|                             | 0.1      | 2.88                                   | 2.49 | 2.28 | 2.14 | 2.05 | 1.98 | 1.93 | 1.88 | 1.85 | 1.82 |
|                             | 0.05     | 4.17                                   | 3.32 | 2.92 | 2.69 | 2.53 | 2.42 | 2.33 | 2.27 | 2.21 | 2.16 |
| 40                          | 0.025    | 5.57                                   | 4.18 | 3.59 | 3.25 | 3.03 | 2.87 | 2.75 | 2.65 | 2.57 | 2.51 |
|                             | 0.01     | 7.56                                   | 5.39 | 4.51 | 4.02 | 3.7  | 3.47 | 3.3  | 3.17 | 3.07 | 2.98 |
|                             | 0.005    | 9.18                                   | 6.35 | 5.24 | 4.62 | 4.23 | 3.95 | 3.74 | 3.58 | 3.45 | 3.34 |
|                             | 0.001    | 13.3                                   | 8.77 | 7.05 | 6.12 | 5.53 | 5.12 | 4.82 | 4.58 | 4.39 | 4.24 |
|                             | 0.25     | 1.36                                   | 1.44 | 1.42 | 1.4  | 1.39 | 1.37 | 1.36 | 1.35 | 1.34 | 1.33 |
|                             | 0.1      | 2.84                                   | 2.44 | 2.23 | 2.09 | 2    | 1.93 | 1.87 | 1.83 | 1.79 | 1.76 |
| 50                          | 0.05     | 4.08                                   | 3.23 | 2.84 | 2.61 | 2.45 | 2.34 | 2.25 | 2.18 | 2.12 | 2.08 |
|                             | 0.025    | 5.42                                   | 4.05 | 3.46 | 3.13 | 2.9  | 2.74 | 2.62 | 2.53 | 2.45 | 2.39 |
|                             | 0.01     | 7.31                                   | 5.18 | 4.31 | 3.83 | 3.51 | 3.29 | 3.12 | 2.99 | 2.89 | 2.8  |
|                             | 0.005    | 8.83                                   | 6.07 | 4.98 | 4.37 | 3.99 | 3.71 | 3.51 | 3.35 | 3.22 | 3.12 |
|                             | 0.001    | 12.6                                   | 8.25 | 6.59 | 5.7  | 5.13 | 4.73 | 4.44 | 4.21 | 4.02 | 3.87 |
|                             | 0.25     | 1.35                                   | 1.43 | 1.41 | 1.39 | 1.37 | 1.36 | 1.34 | 1.33 | 1.32 | 1.31 |
| 100                         | 0.1      | 2.81                                   | 2.41 | 2.2  | 2.06 | 1.97 | 1.9  | 1.84 | 1.8  | 1.76 | 1.73 |
|                             | 0.05     | 4.03                                   | 3.18 | 2.79 | 2.56 | 2.4  | 2.29 | 2.2  | 2.13 | 2.07 | 2.03 |
|                             | 0.025    | 5.34                                   | 3.97 | 3.39 | 3.05 | 2.83 | 2.67 | 2.55 | 2.46 | 2.38 | 2.32 |
|                             | 0.01     | 7.17                                   | 5.06 | 4.2  | 3.72 | 3.41 | 3.19 | 3.02 | 2.89 | 2.78 | 2.7  |
|                             | 0.005    | 8.63                                   | 5.9  | 4.83 | 4.23 | 3.85 | 3.58 | 3.38 | 3.22 | 3.09 | 2.99 |
|                             | 0.001    | 12.2                                   | 7.96 | 6.34 | 5.46 | 4.9  | 4.51 | 4.22 | 4    | 3.82 | 3.67 |
| 200                         | 0.25     | 1.34                                   | 1.41 | 1.39 | 1.37 | 1.35 | 1.33 | 1.32 | 1.3  | 1.29 | 1.28 |
|                             | 0.1      | 2.76                                   | 2.36 | 2.14 | 2    | 1.91 | 1.83 | 1.78 | 1.73 | 1.69 | 1.66 |
|                             | 0.05     | 3.94                                   | 3.09 | 2.7  | 2.46 | 2.31 | 2.19 | 2.1  | 2.03 | 1.97 | 1.93 |
|                             | 0.025    | 5.18                                   | 3.83 | 3.25 | 2.92 | 2.7  | 2.54 | 2.42 | 2.32 | 2.24 | 2.18 |
|                             | 0.01     | 6.9                                    | 4.82 | 3.98 | 3.51 | 3.21 | 2.99 | 2.82 | 2.69 | 2.59 | 2.5  |
|                             | 0.005    | 8.24                                   | 5.59 | 4.54 | 3.96 | 3.59 | 3.33 | 3.13 | 2.97 | 2.85 | 2.74 |
| $\infty$                    | 0.001    | 11.5                                   | 7.41 | 5.86 | 5.02 | 4.48 | 4.11 | 3.83 | 3.61 | 3.44 | 3.3  |
|                             | 0.25     | 1.33                                   | 1.4  | 1.38 | 1.36 | 1.34 | 1.32 | 1.3  | 1.29 | 1.28 | 1.27 |
|                             | 0.1      | 2.73                                   | 2.33 | 2.11 | 1.97 | 1.88 | 1.8  | 1.75 | 1.7  | 1.66 | 1.63 |
|                             | 0.05     | 3.89                                   | 3.04 | 2.65 | 2.42 | 2.26 | 2.14 | 2.06 | 1.98 | 1.93 | 1.88 |
|                             | 0.025    | 5.1                                    | 3.76 | 3.18 | 2.85 | 2.63 | 2.47 | 2.35 | 2.26 | 2.18 | 2.11 |
|                             | 0.01     | 6.76                                   | 4.71 | 3.88 | 3.41 | 3.11 | 2.89 | 2.73 | 2.6  | 2.5  | 2.41 |
| $\infty$                    | 0.005    | 8.06                                   | 5.44 | 4.41 | 3.84 | 3.47 | 3.21 | 3.01 | 2.86 | 2.73 | 2.63 |
|                             | 0.001    | 11.2                                   | 7.15 | 5.63 | 4.81 | 4.29 | 3.92 | 3.65 | 3.43 | 3.26 | 3.12 |
|                             | 0.25     | 1.32                                   | 1.39 | 1.37 | 1.35 | 1.33 | 1.31 | 1.29 | 1.28 | 1.27 | 1.25 |
|                             | 0.1      | 2.71                                   | 2.3  | 2.08 | 1.94 | 1.85 | 1.77 | 1.72 | 1.67 | 1.63 | 1.6  |
|                             | 0.05     | 3.84                                   | 3    | 2.6  | 2.37 | 2.21 | 2.1  | 2.01 | 1.94 | 1.88 | 1.83 |
|                             | 0.025    | 5.02                                   | 3.69 | 3.12 | 2.79 | 2.57 | 2.41 | 2.29 | 2.19 | 2.11 | 2.05 |
| $\infty$                    | 0.01     | 6.63                                   | 4.61 | 3.78 | 3.32 | 3.02 | 2.8  | 2.64 | 2.51 | 2.41 | 2.32 |
|                             | 0.005    | 7.88                                   | 5.3  | 4.28 | 3.72 | 3.35 | 3.09 | 2.9  | 2.74 | 2.62 | 2.52 |
|                             | 0.001    | 10.8                                   | 6.91 | 5.42 | 4.62 | 4.1  | 3.74 | 3.47 | 3.27 | 3.1  | 2.96 |

Table A7, continued. F-distribution

| $\nu_2$ ,<br>denom.<br>d.f. | $\alpha$ | $\nu_1$ , numerator degrees of freedom |      |      |      |      |      |      |      |      |           |
|-----------------------------|----------|----------------------------------------|------|------|------|------|------|------|------|------|-----------|
|                             |          | 15                                     | 20   | 25   | 30   | 40   | 50   | 100  | 200  | 500  | $\infty$  |
| 15                          | 0.25     | 1.43                                   | 1.41 | 1.4  | 1.4  | 1.39 | 1.38 | 1.37 | 1.37 | 1.36 | 1.36      |
|                             | 0.1      | 1.97                                   | 1.92 | 1.89 | 1.87 | 1.85 | 1.83 | 1.79 | 1.77 | 1.76 | 1.76      |
|                             | 0.05     | 2.4                                    | 2.33 | 2.28 | 2.25 | 2.2  | 2.18 | 2.12 | 2.1  | 2.08 | 2.07      |
|                             | 0.025    | 2.86                                   | 2.76 | 2.69 | 2.64 | 2.59 | 2.55 | 2.47 | 2.44 | 2.41 | 2.4       |
|                             | 0.01     | 3.52                                   | 3.37 | 3.28 | 3.21 | 3.13 | 3.08 | 2.98 | 2.92 | 2.89 | 2.87      |
|                             | 0.005    | 4.07                                   | 3.88 | 3.77 | 3.69 | 3.58 | 3.52 | 3.39 | 3.33 | 3.29 | 3.26      |
|                             | 0.001    | 5.54                                   | 5.25 | 5.07 | 4.95 | 4.8  | 4.7  | 4.51 | 4.41 | 4.35 | 4.31      |
| 20                          | 0.25     | 1.37                                   | 1.36 | 1.35 | 1.34 | 1.33 | 1.32 | 1.31 | 1.3  | 1.3  | 1.29      |
|                             | 0.1      | 1.84                                   | 1.79 | 1.76 | 1.74 | 1.71 | 1.69 | 1.65 | 1.63 | 1.62 | 1.61      |
|                             | 0.05     | 2.2                                    | 2.12 | 2.07 | 2.04 | 1.99 | 1.97 | 1.91 | 1.88 | 1.86 | 1.84      |
|                             | 0.025    | 2.57                                   | 2.46 | 2.4  | 2.35 | 2.29 | 2.25 | 2.17 | 2.13 | 2.1  | 2.09      |
|                             | 0.01     | 3.09                                   | 2.94 | 2.84 | 2.78 | 2.69 | 2.64 | 2.54 | 2.48 | 2.44 | 2.42      |
|                             | 0.005    | 3.5                                    | 3.32 | 3.2  | 3.12 | 3.02 | 2.96 | 2.83 | 2.76 | 2.72 | 2.69      |
|                             | 0.001    | 4.56                                   | 4.29 | 4.12 | 4    | 3.86 | 3.77 | 3.58 | 3.48 | 3.42 | 3.38      |
| 25                          | 0.25     | 1.34                                   | 1.33 | 1.31 | 1.31 | 1.29 | 1.29 | 1.27 | 1.26 | 1.26 | 1.25      |
|                             | 0.1      | 1.77                                   | 1.72 | 1.68 | 1.66 | 1.63 | 1.61 | 1.56 | 1.54 | 1.53 | 1.52      |
|                             | 0.05     | 2.09                                   | 2.01 | 1.96 | 1.92 | 1.87 | 1.84 | 1.78 | 1.75 | 1.73 | 1.71      |
|                             | 0.025    | 2.41                                   | 2.3  | 2.23 | 2.18 | 2.12 | 2.08 | 2    | 1.95 | 1.92 | 1.91      |
|                             | 0.01     | 2.85                                   | 2.7  | 2.6  | 2.54 | 2.45 | 2.4  | 2.29 | 2.23 | 2.19 | 2.17      |
|                             | 0.005    | 3.2                                    | 3.01 | 2.9  | 2.82 | 2.72 | 2.65 | 2.52 | 2.45 | 2.41 | 2.38      |
|                             | 0.001    | 4.06                                   | 3.79 | 3.63 | 3.52 | 3.37 | 3.28 | 3.09 | 2.99 | 2.93 | 2.89      |
| 30                          | 0.25     | 1.32                                   | 1.3  | 1.29 | 1.28 | 1.27 | 1.26 | 1.25 | 1.24 | 1.23 | 1.23      |
|                             | 0.1      | 1.72                                   | 1.67 | 1.63 | 1.61 | 1.57 | 1.55 | 1.51 | 1.48 | 1.47 | 1.46      |
|                             | 0.05     | 2.01                                   | 1.93 | 1.88 | 1.84 | 1.79 | 1.76 | 1.7  | 1.66 | 1.64 | 1.62      |
|                             | 0.025    | 2.31                                   | 2.2  | 2.12 | 2.07 | 2.01 | 1.97 | 1.88 | 1.84 | 1.81 | 1.79      |
|                             | 0.01     | 2.7                                    | 2.55 | 2.45 | 2.39 | 2.3  | 2.25 | 2.13 | 2.07 | 2.03 | 2.01      |
|                             | 0.005    | 3.01                                   | 2.82 | 2.71 | 2.63 | 2.52 | 2.46 | 2.32 | 2.25 | 2.21 | 2.18      |
|                             | 0.001    | 3.75                                   | 3.49 | 3.33 | 3.22 | 3.07 | 2.98 | 2.79 | 2.69 | 2.63 | 2.59      |
| 40                          | 0.25     | 1.3                                    | 1.28 | 1.26 | 1.25 | 1.24 | 1.23 | 1.21 | 1.2  | 1.19 | 1.19      |
|                             | 0.1      | 1.66                                   | 1.61 | 1.57 | 1.54 | 1.51 | 1.48 | 1.43 | 1.41 | 1.39 | 1.38      |
|                             | 0.05     | 1.92                                   | 1.84 | 1.78 | 1.74 | 1.69 | 1.66 | 1.59 | 1.55 | 1.53 | 1.51      |
|                             | 0.025    | 2.18                                   | 2.07 | 1.99 | 1.94 | 1.88 | 1.83 | 1.74 | 1.69 | 1.66 | 1.64      |
|                             | 0.01     | 2.52                                   | 2.37 | 2.27 | 2.2  | 2.11 | 2.06 | 1.94 | 1.87 | 1.83 | 1.8       |
|                             | 0.005    | 2.78                                   | 2.6  | 2.48 | 2.4  | 2.3  | 2.23 | 2.09 | 2.01 | 1.96 | 1.93      |
|                             | 0.001    | 3.4                                    | 3.14 | 2.98 | 2.87 | 2.73 | 2.64 | 2.44 | 2.34 | 2.28 | 2.23      |
| 50                          | 0.25     | 1.28                                   | 1.26 | 1.25 | 1.23 | 1.22 | 1.21 | 1.19 | 1.18 | 1.17 | 1.16      |
|                             | 0.1      | 1.63                                   | 1.57 | 1.53 | 1.5  | 1.46 | 1.44 | 1.39 | 1.36 | 1.34 | 1.33      |
|                             | 0.05     | 1.87                                   | 1.78 | 1.73 | 1.69 | 1.63 | 1.6  | 1.52 | 1.48 | 1.46 | 1.44      |
|                             | 0.025    | 2.11                                   | 1.99 | 1.92 | 1.87 | 1.8  | 1.75 | 1.66 | 1.6  | 1.57 | 1.55      |
|                             | 0.01     | 2.42                                   | 2.27 | 2.17 | 2.1  | 2.01 | 1.95 | 1.82 | 1.76 | 1.71 | 1.68      |
|                             | 0.005    | 2.65                                   | 2.47 | 2.35 | 2.27 | 2.16 | 2.1  | 1.95 | 1.87 | 1.82 | 1.79      |
|                             | 0.001    | 3.2                                    | 2.95 | 2.79 | 2.68 | 2.53 | 2.44 | 2.25 | 2.14 | 2.07 | 2.03      |
| 100                         | 0.25     | 1.25                                   | 1.23 | 1.21 | 1.2  | 1.18 | 1.17 | 1.14 | 1.13 | 1.12 | 1.11      |
|                             | 0.1      | 1.56                                   | 1.49 | 1.45 | 1.42 | 1.38 | 1.35 | 1.29 | 1.26 | 1.23 | 1.21      |
|                             | 0.05     | 1.77                                   | 1.68 | 1.62 | 1.57 | 1.52 | 1.48 | 1.39 | 1.34 | 1.31 | 1.28      |
|                             | 0.025    | 1.97                                   | 1.85 | 1.77 | 1.71 | 1.64 | 1.59 | 1.48 | 1.42 | 1.38 | 1.35      |
|                             | 0.01     | 2.22                                   | 2.07 | 1.97 | 1.89 | 1.8  | 1.74 | 1.6  | 1.52 | 1.47 | 1.43      |
|                             | 0.005    | 2.41                                   | 2.23 | 2.11 | 2.02 | 1.91 | 1.84 | 1.68 | 1.59 | 1.53 | 1.49      |
|                             | 0.001    | 2.84                                   | 2.59 | 2.43 | 2.32 | 2.17 | 2.08 | 1.87 | 1.75 | 1.67 | 1.62      |
| 200                         | 0.25     | 1.23                                   | 1.21 | 1.19 | 1.18 | 1.16 | 1.15 | 1.12 | 1.1  | 1.09 | 1.07      |
|                             | 0.1      | 1.52                                   | 1.46 | 1.41 | 1.38 | 1.34 | 1.31 | 1.24 | 1.2  | 1.17 | 1.14      |
|                             | 0.05     | 1.72                                   | 1.62 | 1.56 | 1.52 | 1.46 | 1.41 | 1.32 | 1.26 | 1.22 | 1.19      |
|                             | 0.025    | 1.9                                    | 1.78 | 1.7  | 1.64 | 1.56 | 1.51 | 1.39 | 1.32 | 1.27 | 1.23      |
|                             | 0.01     | 2.13                                   | 1.97 | 1.87 | 1.79 | 1.69 | 1.63 | 1.48 | 1.39 | 1.33 | 1.28      |
|                             | 0.005    | 2.3                                    | 2.11 | 1.99 | 1.91 | 1.79 | 1.71 | 1.54 | 1.44 | 1.37 | 1.31      |
|                             | 0.001    | 2.67                                   | 2.42 | 2.26 | 2.15 | 2    | 1.9  | 1.68 | 1.55 | 1.46 | 1.39      |
| $\infty$                    | 0.25     | 1.22                                   | 1.19 | 1.17 | 1.16 | 1.14 | 1.13 | 1.09 | 1.07 | 1.04 | Undefined |
|                             | 0.1      | 1.49                                   | 1.42 | 1.38 | 1.34 | 1.3  | 1.26 | 1.18 | 1.13 | 1.08 |           |
|                             | 0.05     | 1.67                                   | 1.57 | 1.51 | 1.46 | 1.39 | 1.35 | 1.24 | 1.17 | 1.11 |           |
|                             | 0.025    | 1.83                                   | 1.71 | 1.63 | 1.57 | 1.48 | 1.43 | 1.3  | 1.21 | 1.13 |           |
|                             | 0.01     | 2.04                                   | 1.88 | 1.77 | 1.7  | 1.59 | 1.52 | 1.36 | 1.25 | 1.15 |           |
|                             | 0.005    | 2.19                                   | 2    | 1.88 | 1.79 | 1.67 | 1.59 | 1.4  | 1.28 | 1.17 |           |
|                             | 0.001    | 2.51                                   | 2.27 | 2.1  | 1.99 | 1.84 | 1.73 | 1.49 | 1.34 | 1.21 |           |

**Table A8. Table of Critical Values for the Wilcoxon Signed Rank Test**

For the left-tail test, the table gives the largest integer  $w$  such that  $\mathbf{P}\{W \leq w \mid H_0\} \leq \alpha$ .

For the right-tail test, the table gives the smallest integer  $w$  such that  $\mathbf{P}\{W \geq w \mid H_0\} \leq \alpha$ .

A missing table entry means that such an integer does not exist among possible values of  $W$ .

| $n$ | $\alpha$ , left-tail probability for the left-tail test |       |       |       |       |       |       | $\alpha$ , right-tail probability for the right-tail test |       |       |       |       |       |       |
|-----|---------------------------------------------------------|-------|-------|-------|-------|-------|-------|-----------------------------------------------------------|-------|-------|-------|-------|-------|-------|
|     | 0.001                                                   | 0.005 | 0.010 | 0.025 | 0.050 | 0.100 | 0.200 | 0.200                                                     | 0.100 | 0.050 | 0.025 | 0.010 | 0.005 | 0.001 |
| 1   | —                                                       | —     | —     | —     | —     | —     | —     | —                                                         | —     | —     | —     | —     | —     | —     |
| 2   | —                                                       | —     | —     | —     | —     | —     | —     | —                                                         | —     | —     | —     | —     | —     | —     |
| 3   | —                                                       | —     | —     | —     | —     | —     | 0     | 6                                                         | —     | —     | —     | —     | —     | —     |
| 4   | —                                                       | —     | —     | —     | —     | 0     | 2     | 8                                                         | 10    | —     | —     | —     | —     | —     |
| 5   | —                                                       | —     | —     | —     | 0     | 2     | 3     | 12                                                        | 13    | 15    | —     | —     | —     | —     |
| 6   | —                                                       | —     | —     | 0     | 2     | 3     | 5     | 16                                                        | 18    | 19    | 21    | —     | —     | —     |
| 7   | —                                                       | —     | 0     | 2     | 3     | 5     | 8     | 20                                                        | 23    | 25    | 26    | 28    | —     | —     |
| 8   | —                                                       | 0     | 1     | 3     | 5     | 8     | 11    | 25                                                        | 28    | 31    | 33    | 35    | 36    | —     |
| 9   | —                                                       | 1     | 3     | 5     | 8     | 10    | 14    | 31                                                        | 35    | 37    | 40    | 42    | 44    | —     |
| 10  | 0                                                       | 3     | 5     | 8     | 10    | 14    | 18    | 37                                                        | 41    | 45    | 47    | 50    | 52    | 55    |
| 11  | 1                                                       | 5     | 7     | 10    | 13    | 17    | 22    | 44                                                        | 49    | 53    | 56    | 59    | 61    | 65    |
| 12  | 2                                                       | 7     | 9     | 13    | 17    | 21    | 27    | 51                                                        | 57    | 61    | 65    | 69    | 71    | 76    |
| 13  | 4                                                       | 9     | 12    | 17    | 21    | 26    | 32    | 59                                                        | 65    | 70    | 74    | 79    | 82    | 87    |
| 14  | 6                                                       | 12    | 15    | 21    | 25    | 31    | 38    | 67                                                        | 74    | 80    | 84    | 90    | 93    | 99    |
| 15  | 8                                                       | 15    | 19    | 25    | 30    | 36    | 44    | 76                                                        | 84    | 90    | 95    | 101   | 105   | 112   |
| 16  | 11                                                      | 19    | 23    | 29    | 35    | 42    | 50    | 86                                                        | 94    | 101   | 107   | 113   | 117   | 125   |
| 17  | 14                                                      | 23    | 27    | 34    | 41    | 48    | 57    | 96                                                        | 105   | 112   | 119   | 126   | 130   | 139   |
| 18  | 18                                                      | 27    | 32    | 40    | 47    | 55    | 65    | 106                                                       | 116   | 124   | 131   | 139   | 144   | 153   |
| 19  | 21                                                      | 32    | 37    | 46    | 53    | 62    | 73    | 117                                                       | 128   | 137   | 144   | 153   | 158   | 169   |
| 20  | 26                                                      | 37    | 43    | 52    | 60    | 69    | 81    | 129                                                       | 141   | 150   | 158   | 167   | 173   | 184   |
| 21  | 30                                                      | 42    | 49    | 58    | 67    | 77    | 90    | 141                                                       | 154   | 164   | 173   | 182   | 189   | 201   |
| 22  | 35                                                      | 48    | 55    | 65    | 75    | 86    | 99    | 154                                                       | 167   | 178   | 188   | 198   | 205   | 218   |
| 23  | 40                                                      | 54    | 62    | 73    | 83    | 94    | 109   | 167                                                       | 182   | 193   | 203   | 214   | 222   | 236   |
| 24  | 45                                                      | 61    | 69    | 81    | 91    | 104   | 119   | 181                                                       | 196   | 209   | 219   | 231   | 239   | 255   |
| 25  | 51                                                      | 68    | 76    | 89    | 100   | 113   | 130   | 195                                                       | 212   | 225   | 236   | 249   | 257   | 274   |
| 26  | 58                                                      | 75    | 84    | 98    | 110   | 124   | 141   | 210                                                       | 227   | 241   | 253   | 267   | 276   | 293   |
| 27  | 64                                                      | 83    | 92    | 107   | 119   | 134   | 153   | 225                                                       | 244   | 259   | 271   | 286   | 295   | 314   |
| 28  | 71                                                      | 91    | 101   | 116   | 130   | 145   | 165   | 241                                                       | 261   | 276   | 290   | 305   | 315   | 335   |
| 29  | 79                                                      | 100   | 110   | 126   | 140   | 157   | 177   | 258                                                       | 278   | 295   | 309   | 325   | 335   | 356   |
| 30  | 86                                                      | 109   | 120   | 137   | 151   | 169   | 190   | 275                                                       | 296   | 314   | 328   | 345   | 356   | 379   |

**Table A9. Critical Values for the Mann-Whitney-Wilcoxon Rank-Sum Test**

For the left-tail test, the table gives the largest integer  $u$  such that  $\mathbf{P}\{U \leq u \mid H_0\} \leq \alpha$ .

For the right-tail test, the table gives the smallest integer  $u$  such that  $\mathbf{P}\{U \geq u \mid H_0\} \leq \alpha$ .

A missing table entry means that such an integer does not exist among possible values of  $U$ .

| $n_1$ | $n_2$ | $\alpha$ , left-tail probability for the left-tail test<br>$H_A : X$ is stochastically smaller than $Y$ |       |       |       |       |       |       | $\alpha$ , right-tail probability for the right-tail test<br>$H_A : X$ is stochastically larger than $Y$ |       |       |       |       |       |       |
|-------|-------|---------------------------------------------------------------------------------------------------------|-------|-------|-------|-------|-------|-------|----------------------------------------------------------------------------------------------------------|-------|-------|-------|-------|-------|-------|
|       |       | 0.001                                                                                                   | 0.005 | 0.010 | 0.025 | 0.050 | 0.100 | 0.200 | 0.200                                                                                                    | 0.100 | 0.050 | 0.025 | 0.010 | 0.005 | 0.001 |
| 3     | 2     | —                                                                                                       | —     | —     | —     | —     | 6     | 7     | 11                                                                                                       | 12    | —     | —     | —     | —     | —     |
| 3     | 3     | —                                                                                                       | —     | —     | —     | 6     | 7     | 8     | 13                                                                                                       | 14    | 15    | —     | —     | —     | —     |
| 3     | 4     | —                                                                                                       | —     | —     | —     | 6     | 7     | 9     | 15                                                                                                       | 17    | 18    | —     | —     | —     | —     |
| 3     | 5     | —                                                                                                       | —     | —     | 6     | 7     | 8     | 10    | 17                                                                                                       | 19    | 20    | 21    | —     | —     | —     |
| 3     | 6     | —                                                                                                       | —     | —     | 7     | 8     | 9     | 11    | 19                                                                                                       | 21    | 22    | 23    | —     | —     | —     |
| 3     | 7     | —                                                                                                       | —     | 6     | 7     | 8     | 10    | 12    | 21                                                                                                       | 23    | 25    | 26    | 27    | —     | —     |
| 3     | 8     | —                                                                                                       | —     | 6     | 8     | 9     | 11    | 13    | 23                                                                                                       | 25    | 27    | 28    | 30    | —     | —     |
| 3     | 9     | —                                                                                                       | 6     | 7     | 8     | 10    | 11    | 14    | 25                                                                                                       | 28    | 29    | 31    | 32    | 33    | —     |
| 3     | 10    | —                                                                                                       | 6     | 7     | 9     | 10    | 12    | 15    | 27                                                                                                       | 30    | 32    | 33    | 35    | 36    | —     |
| 3     | 11    | —                                                                                                       | 6     | 7     | 9     | 11    | 13    | 16    | 29                                                                                                       | 32    | 34    | 36    | 38    | 39    | —     |
| 3     | 12    | —                                                                                                       | 7     | 8     | 10    | 11    | 14    | 17    | 31                                                                                                       | 34    | 37    | 38    | 40    | 41    | —     |
| 4     | 2     | —                                                                                                       | —     | —     | —     | —     | 10    | 11    | 17                                                                                                       | 18    | —     | —     | —     | —     | —     |
| 4     | 3     | —                                                                                                       | —     | —     | —     | 10    | 11    | 13    | 19                                                                                                       | 21    | 22    | —     | —     | —     | —     |
| 4     | 4     | —                                                                                                       | —     | —     | 10    | 11    | 13    | 14    | 22                                                                                                       | 23    | 25    | 26    | —     | —     | —     |
| 4     | 5     | —                                                                                                       | —     | 10    | 11    | 12    | 14    | 15    | 25                                                                                                       | 26    | 28    | 29    | 30    | —     | —     |
| 4     | 6     | —                                                                                                       | 10    | 11    | 12    | 13    | 15    | 17    | 27                                                                                                       | 29    | 31    | 32    | 33    | 34    | —     |
| 4     | 7     | —                                                                                                       | 10    | 11    | 13    | 14    | 16    | 18    | 30                                                                                                       | 32    | 34    | 35    | 37    | 38    | —     |
| 4     | 8     | —                                                                                                       | 11    | 12    | 14    | 15    | 17    | 20    | 32                                                                                                       | 35    | 37    | 38    | 40    | 41    | —     |
| 4     | 9     | —                                                                                                       | 11    | 13    | 14    | 16    | 19    | 21    | 35                                                                                                       | 37    | 40    | 42    | 43    | 45    | —     |
| 4     | 10    | 10                                                                                                      | 12    | 13    | 15    | 17    | 20    | 23    | 37                                                                                                       | 40    | 43    | 45    | 47    | 48    | 50    |
| 4     | 11    | 10                                                                                                      | 12    | 14    | 16    | 18    | 21    | 24    | 40                                                                                                       | 43    | 46    | 48    | 50    | 52    | 54    |
| 4     | 12    | 10                                                                                                      | 13    | 15    | 17    | 19    | 22    | 26    | 42                                                                                                       | 46    | 49    | 51    | 53    | 55    | 58    |
| 5     | 2     | —                                                                                                       | —     | —     | —     | 15    | 16    | 17    | 23                                                                                                       | 24    | 25    | —     | —     | —     | —     |
| 5     | 3     | —                                                                                                       | —     | —     | 15    | 16    | 17    | 19    | 26                                                                                                       | 28    | 29    | 30    | —     | —     | —     |
| 5     | 4     | —                                                                                                       | —     | 15    | 16    | 17    | 19    | 20    | 30                                                                                                       | 31    | 33    | 34    | 35    | —     | —     |
| 5     | 5     | —                                                                                                       | 15    | 16    | 17    | 19    | 20    | 22    | 33                                                                                                       | 35    | 36    | 38    | 39    | 40    | —     |
| 5     | 6     | —                                                                                                       | 16    | 17    | 18    | 20    | 22    | 24    | 36                                                                                                       | 38    | 40    | 42    | 43    | 44    | —     |
| 5     | 7     | —                                                                                                       | 16    | 18    | 20    | 21    | 23    | 26    | 39                                                                                                       | 42    | 44    | 45    | 47    | 49    | —     |
| 5     | 8     | 15                                                                                                      | 17    | 19    | 21    | 23    | 25    | 28    | 42                                                                                                       | 45    | 47    | 49    | 51    | 53    | 55    |
| 5     | 9     | 16                                                                                                      | 18    | 20    | 22    | 24    | 27    | 30    | 45                                                                                                       | 48    | 51    | 53    | 55    | 57    | 59    |
| 5     | 10    | 16                                                                                                      | 19    | 21    | 23    | 26    | 28    | 32    | 48                                                                                                       | 52    | 54    | 57    | 59    | 61    | 64    |
| 5     | 11    | 17                                                                                                      | 20    | 22    | 24    | 27    | 30    | 34    | 51                                                                                                       | 55    | 58    | 61    | 63    | 65    | 68    |
| 5     | 12    | 17                                                                                                      | 21    | 23    | 26    | 28    | 32    | 36    | 54                                                                                                       | 58    | 62    | 64    | 67    | 69    | 73    |
| 6     | 2     | —                                                                                                       | —     | —     | —     | 21    | 22    | 23    | 31                                                                                                       | 32    | 33    | —     | —     | —     | —     |
| 6     | 3     | —                                                                                                       | —     | —     | 22    | 23    | 24    | 26    | 34                                                                                                       | 36    | 37    | 38    | —     | —     | —     |
| 6     | 4     | —                                                                                                       | 21    | 22    | 23    | 24    | 26    | 28    | 38                                                                                                       | 40    | 42    | 43    | 44    | 45    | —     |
| 6     | 5     | —                                                                                                       | 22    | 23    | 24    | 26    | 28    | 30    | 42                                                                                                       | 44    | 46    | 48    | 49    | 50    | —     |
| 6     | 6     | —                                                                                                       | 23    | 24    | 26    | 28    | 30    | 33    | 45                                                                                                       | 48    | 50    | 52    | 54    | 55    | —     |
| 6     | 7     | 21                                                                                                      | 24    | 25    | 27    | 29    | 32    | 35    | 49                                                                                                       | 52    | 55    | 57    | 59    | 60    | 63    |
| 6     | 8     | 22                                                                                                      | 25    | 27    | 29    | 31    | 34    | 37    | 53                                                                                                       | 56    | 59    | 61    | 63    | 65    | 68    |
| 6     | 9     | 23                                                                                                      | 26    | 28    | 31    | 33    | 36    | 40    | 56                                                                                                       | 60    | 63    | 65    | 68    | 70    | 73    |
| 6     | 10    | 24                                                                                                      | 27    | 29    | 32    | 35    | 38    | 42    | 60                                                                                                       | 64    | 67    | 70    | 73    | 75    | 78    |
| 6     | 11    | 25                                                                                                      | 28    | 30    | 34    | 37    | 40    | 44    | 64                                                                                                       | 68    | 71    | 74    | 78    | 80    | 83    |
| 6     | 12    | 25                                                                                                      | 30    | 32    | 35    | 38    | 42    | 47    | 67                                                                                                       | 72    | 76    | 79    | 82    | 84    | 89    |
| 7     | 2     | —                                                                                                       | —     | —     | —     | 28    | 29    | 31    | 39                                                                                                       | 41    | 42    | —     | —     | —     | —     |
| 7     | 3     | —                                                                                                       | —     | 28    | 29    | 30    | 32    | 34    | 43                                                                                                       | 45    | 47    | 48    | 49    | —     | —     |
| 7     | 4     | —                                                                                                       | 28    | 29    | 31    | 32    | 34    | 36    | 48                                                                                                       | 50    | 52    | 53    | 55    | 56    | —     |
| 7     | 5     | —                                                                                                       | 29    | 31    | 33    | 34    | 36    | 39    | 52                                                                                                       | 55    | 57    | 58    | 60    | 62    | —     |
| 7     | 6     | 28                                                                                                      | 31    | 32    | 34    | 36    | 39    | 42    | 56                                                                                                       | 59    | 62    | 64    | 66    | 67    | 70    |
| 7     | 7     | 29                                                                                                      | 32    | 34    | 36    | 39    | 41    | 45    | 60                                                                                                       | 64    | 66    | 69    | 71    | 73    | 76    |
| 7     | 8     | 30                                                                                                      | 34    | 35    | 38    | 41    | 44    | 48    | 64                                                                                                       | 68    | 71    | 74    | 77    | 78    | 82    |
| 7     | 9     | 31                                                                                                      | 35    | 37    | 40    | 43    | 46    | 50    | 69                                                                                                       | 73    | 76    | 79    | 82    | 84    | 88    |
| 7     | 10    | 33                                                                                                      | 37    | 39    | 42    | 45    | 49    | 53    | 73                                                                                                       | 77    | 81    | 84    | 87    | 89    | 93    |
| 7     | 11    | 34                                                                                                      | 38    | 40    | 44    | 47    | 51    | 56    | 77                                                                                                       | 82    | 86    | 89    | 93    | 95    | 99    |
| 7     | 12    | 35                                                                                                      | 40    | 42    | 46    | 49    | 54    | 59    | 81                                                                                                       | 86    | 91    | 94    | 98    | 100   | 105   |



**Table A9, continued. Critical Values for the Mann-Whitney-Wilcoxon Rank-Sum Test**

For the left-tail test, the table gives the largest integer  $u$  such that  $\mathbf{P}\{U \leq u \mid H_0\} \leq \alpha$ .

For the right-tail test, the table gives the smallest integer  $u$  such that  $\mathbf{P}\{U \geq u \mid H_0\} \leq \alpha$ .

A missing table entry means that such an integer does not exist among possible values of  $U$ .

| $n_1$ | $n_2$ | $\alpha$ , left-tail probability for the left-tail test<br>$H_A : X$ is stochastically smaller than $Y$ |       |       |       |       |       |       | $\alpha$ , right-tail probability for the right-tail test<br>$H_A : X$ is stochastically larger than $Y$ |       |       |       |       |       |       |
|-------|-------|---------------------------------------------------------------------------------------------------------|-------|-------|-------|-------|-------|-------|----------------------------------------------------------------------------------------------------------|-------|-------|-------|-------|-------|-------|
|       |       | 0.001                                                                                                   | 0.005 | 0.010 | 0.025 | 0.050 | 0.100 | 0.200 | 0.200                                                                                                    | 0.100 | 0.050 | 0.025 | 0.010 | 0.005 | 0.001 |
| 8     | 2     | —                                                                                                       | —     | —     | 36    | 37    | 38    | 40    | 48                                                                                                       | 50    | 51    | 52    | —     | —     | —     |
| 8     | 3     | —                                                                                                       | —     | 36    | 38    | 39    | 41    | 43    | 53                                                                                                       | 55    | 57    | 58    | 60    | —     | —     |
| 8     | 4     | —                                                                                                       | 37    | 38    | 40    | 41    | 43    | 46    | 58                                                                                                       | 61    | 63    | 64    | 66    | 67    | —     |
| 8     | 5     | 36                                                                                                      | 38    | 40    | 42    | 44    | 46    | 49    | 63                                                                                                       | 66    | 68    | 70    | 72    | 74    | 76    |
| 8     | 6     | 37                                                                                                      | 40    | 42    | 44    | 46    | 49    | 52    | 68                                                                                                       | 71    | 74    | 76    | 78    | 80    | 83    |
| 8     | 7     | 38                                                                                                      | 42    | 43    | 46    | 49    | 52    | 56    | 72                                                                                                       | 76    | 79    | 82    | 85    | 86    | 90    |
| 8     | 8     | 40                                                                                                      | 43    | 45    | 49    | 51    | 55    | 59    | 77                                                                                                       | 81    | 85    | 87    | 91    | 93    | 96    |
| 8     | 9     | 41                                                                                                      | 45    | 47    | 51    | 54    | 58    | 62    | 82                                                                                                       | 86    | 90    | 93    | 97    | 99    | 103   |
| 8     | 10    | 42                                                                                                      | 47    | 49    | 53    | 56    | 60    | 65    | 87                                                                                                       | 92    | 96    | 99    | 103   | 105   | 110   |
| 8     | 11    | 44                                                                                                      | 49    | 51    | 55    | 59    | 63    | 69    | 91                                                                                                       | 97    | 101   | 105   | 109   | 111   | 116   |
| 8     | 12    | 45                                                                                                      | 51    | 53    | 58    | 62    | 66    | 72    | 96                                                                                                       | 102   | 106   | 110   | 115   | 117   | 123   |
| 9     | 2     | —                                                                                                       | —     | —     | 45    | 46    | 47    | 49    | 59                                                                                                       | 61    | 62    | 63    | —     | —     | —     |
| 9     | 3     | —                                                                                                       | 45    | 46    | 47    | 49    | 50    | 53    | 64                                                                                                       | 67    | 68    | 70    | 71    | 72    | —     |
| 9     | 4     | —                                                                                                       | 46    | 48    | 49    | 51    | 54    | 56    | 70                                                                                                       | 72    | 75    | 77    | 78    | 80    | —     |
| 9     | 5     | 46                                                                                                      | 48    | 50    | 52    | 54    | 57    | 60    | 75                                                                                                       | 78    | 81    | 83    | 85    | 87    | 89    |
| 9     | 6     | 47                                                                                                      | 50    | 52    | 55    | 57    | 60    | 64    | 80                                                                                                       | 84    | 87    | 89    | 92    | 94    | 97    |
| 9     | 7     | 48                                                                                                      | 52    | 54    | 57    | 60    | 63    | 67    | 86                                                                                                       | 90    | 93    | 96    | 99    | 101   | 105   |
| 9     | 8     | 50                                                                                                      | 54    | 56    | 60    | 63    | 67    | 71    | 91                                                                                                       | 95    | 99    | 102   | 106   | 108   | 112   |
| 9     | 9     | 52                                                                                                      | 56    | 59    | 62    | 66    | 70    | 75    | 96                                                                                                       | 101   | 105   | 109   | 112   | 115   | 119   |
| 9     | 10    | 53                                                                                                      | 58    | 61    | 65    | 69    | 73    | 78    | 102                                                                                                      | 107   | 111   | 115   | 119   | 122   | 127   |
| 9     | 11    | 55                                                                                                      | 61    | 63    | 68    | 72    | 76    | 82    | 107                                                                                                      | 113   | 117   | 121   | 126   | 128   | 134   |
| 9     | 12    | 57                                                                                                      | 63    | 66    | 71    | 75    | 80    | 86    | 112                                                                                                      | 118   | 123   | 127   | 132   | 135   | 141   |
| 10    | 2     | —                                                                                                       | —     | —     | 55    | 56    | 58    | 60    | 70                                                                                                       | 72    | 74    | 75    | —     | —     | —     |
| 10    | 3     | —                                                                                                       | 55    | 56    | 58    | 59    | 61    | 64    | 76                                                                                                       | 79    | 81    | 82    | 84    | 85    | —     |
| 10    | 4     | 55                                                                                                      | 57    | 58    | 60    | 62    | 65    | 68    | 82                                                                                                       | 85    | 88    | 90    | 92    | 93    | 95    |
| 10    | 5     | 56                                                                                                      | 59    | 61    | 63    | 66    | 68    | 72    | 88                                                                                                       | 92    | 94    | 97    | 99    | 101   | 104   |
| 10    | 6     | 58                                                                                                      | 61    | 63    | 66    | 69    | 72    | 76    | 94                                                                                                       | 98    | 101   | 104   | 107   | 109   | 112   |
| 10    | 7     | 60                                                                                                      | 64    | 66    | 69    | 72    | 76    | 80    | 100                                                                                                      | 104   | 108   | 111   | 114   | 116   | 120   |
| 10    | 8     | 61                                                                                                      | 66    | 68    | 72    | 75    | 79    | 84    | 106                                                                                                      | 111   | 115   | 118   | 122   | 124   | 129   |
| 10    | 9     | 63                                                                                                      | 68    | 71    | 75    | 79    | 83    | 88    | 112                                                                                                      | 117   | 121   | 125   | 129   | 132   | 137   |
| 10    | 10    | 65                                                                                                      | 71    | 74    | 78    | 82    | 87    | 93    | 117                                                                                                      | 123   | 128   | 132   | 136   | 139   | 145   |
| 10    | 11    | 67                                                                                                      | 73    | 77    | 81    | 86    | 91    | 97    | 123                                                                                                      | 129   | 134   | 139   | 143   | 147   | 153   |
| 10    | 12    | 69                                                                                                      | 76    | 79    | 84    | 89    | 94    | 101   | 129                                                                                                      | 136   | 141   | 146   | 151   | 154   | 161   |
| 11    | 2     | —                                                                                                       | —     | —     | 66    | 67    | 69    | 71    | 83                                                                                                       | 85    | 87    | 88    | —     | —     | —     |
| 11    | 3     | —                                                                                                       | 66    | 67    | 69    | 71    | 73    | 76    | 89                                                                                                       | 92    | 94    | 96    | 98    | 99    | —     |
| 11    | 4     | 66                                                                                                      | 68    | 70    | 72    | 74    | 77    | 80    | 96                                                                                                       | 99    | 102   | 104   | 106   | 108   | 110   |
| 11    | 5     | 68                                                                                                      | 71    | 73    | 75    | 78    | 81    | 85    | 102                                                                                                      | 106   | 109   | 112   | 114   | 116   | 119   |
| 11    | 6     | 70                                                                                                      | 73    | 75    | 79    | 82    | 85    | 89    | 109                                                                                                      | 113   | 116   | 119   | 123   | 125   | 128   |
| 11    | 7     | 72                                                                                                      | 76    | 78    | 82    | 85    | 89    | 94    | 115                                                                                                      | 120   | 124   | 127   | 131   | 133   | 137   |
| 11    | 8     | 74                                                                                                      | 79    | 81    | 85    | 89    | 93    | 99    | 121                                                                                                      | 127   | 131   | 135   | 139   | 141   | 146   |
| 11    | 9     | 76                                                                                                      | 82    | 84    | 89    | 93    | 97    | 103   | 128                                                                                                      | 134   | 138   | 142   | 147   | 149   | 155   |
| 11    | 10    | 78                                                                                                      | 84    | 88    | 92    | 97    | 102   | 108   | 134                                                                                                      | 140   | 145   | 150   | 154   | 158   | 164   |
| 11    | 11    | 81                                                                                                      | 87    | 91    | 96    | 100   | 106   | 112   | 141                                                                                                      | 147   | 153   | 157   | 162   | 166   | 172   |
| 11    | 12    | 83                                                                                                      | 90    | 94    | 99    | 104   | 110   | 117   | 147                                                                                                      | 154   | 160   | 165   | 170   | 174   | 181   |
| 12    | 2     | —                                                                                                       | —     | —     | 79    | 80    | 82    | 84    | 96                                                                                                       | 98    | 100   | 101   | —     | —     | —     |
| 12    | 3     | —                                                                                                       | 79    | 80    | 82    | 83    | 86    | 89    | 103                                                                                                      | 106   | 109   | 110   | 112   | 113   | —     |
| 12    | 4     | 78                                                                                                      | 81    | 83    | 85    | 87    | 90    | 94    | 110                                                                                                      | 114   | 117   | 119   | 121   | 123   | 126   |
| 12    | 5     | 80                                                                                                      | 84    | 86    | 89    | 91    | 95    | 99    | 117                                                                                                      | 121   | 125   | 127   | 130   | 132   | 136   |
| 12    | 6     | 82                                                                                                      | 87    | 89    | 92    | 95    | 99    | 104   | 124                                                                                                      | 129   | 133   | 136   | 139   | 141   | 146   |
| 12    | 7     | 85                                                                                                      | 90    | 92    | 96    | 99    | 104   | 109   | 131                                                                                                      | 136   | 141   | 144   | 148   | 150   | 155   |
| 12    | 8     | 87                                                                                                      | 93    | 95    | 100   | 104   | 108   | 114   | 138                                                                                                      | 144   | 148   | 152   | 157   | 159   | 165   |
| 12    | 9     | 90                                                                                                      | 96    | 99    | 104   | 108   | 113   | 119   | 145                                                                                                      | 151   | 156   | 160   | 165   | 168   | 174   |
| 12    | 10    | 92                                                                                                      | 99    | 102   | 107   | 112   | 117   | 124   | 152                                                                                                      | 159   | 164   | 169   | 174   | 177   | 184   |
| 12    | 11    | 95                                                                                                      | 102   | 106   | 111   | 116   | 122   | 129   | 159                                                                                                      | 166   | 172   | 177   | 182   | 186   | 193   |
| 12    | 12    | 98                                                                                                      | 105   | 109   | 115   | 120   | 127   | 134   | 166                                                                                                      | 173   | 180   | 185   | 191   | 195   | 202   |